

Carrier-Free Nanoassembly Suppresses Phase Separation via Ribosome-Inspired Crowding Control for Enhanced Chemo-Immunotherapy

Xiaoguang Wang, Lisha Cai, Xiaoxiao Zheng, Li Zheng, Hao Liu, Baoming Wang, Liqiang Hu, Shangzhi Xie, Shufen Zhang, Xuemei Lu, Jiayan Mao, Yiliang Lin, Xiang Gao, Jian Li, Xiaoling Xu,* and Wei Chen*



Cite This: *ACS Nano* 2026, 20, 20471–20488



Read Online

ACCESS |

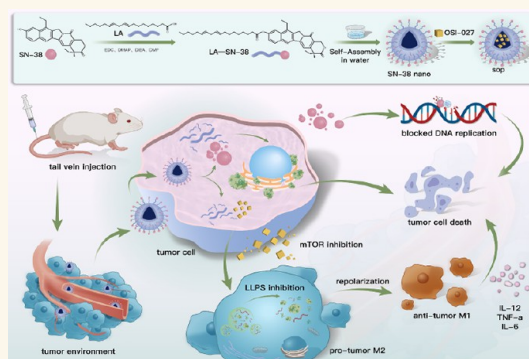
Metrics & More

Article Recommendations

Supporting Information

ABSTRACT: Chemotherapy can induce tumor cell death and stimulate immunity, yet its efficacy is often compromised by the immunosuppressive tumor microenvironment dominated by M2-like tumor-associated macrophages (TAMs). Redirecting TAMs toward an M1-like phenotype through mammalian target of rapamycin (mTOR)-mediated metabolic modulation has therefore emerged as a promising therapeutic strategy. To this end, we developed a carrier-free nanoassembly (SOP) composed of the mTOR inhibitor OSI-027 and the chemotherapeutic agent SN-38. SOP harnesses ribosome-inspired crowding control to suppress liquid–liquid phase separation (LLPS), thereby facilitating M2 reprogramming and enhancing antitumor efficacy. Following intravenous administration, SOP accumulated in tumor tissues with 2-fold greater efficiency than free drugs. Once internalized, SOP released SN-38 to induce tumor cell death and immunogenic responses, while OSI-027 simultaneously inhibited mTORC1/2 signaling and reduced ribosome abundance—key crowding agents driving LLPS. The resulting disruption of LLPS promoted TAM repolarization toward M1, establishing a synergistic interplay between OSI-027 and SN-38. This dual action translated into robust therapeutic outcomes across diverse models, including cell-derived xenografts, patient-derived xenografts, and KPC (LSL-KrasG12D/+; LSL-Trp53R172H/+; Pdx-1-Cre) mice. Moreover, SOP suppressed resistance-related gene expression, overcoming chemotherapy resistance. In summary, this carrier-free nanoassembly not only reprograms the immunosuppressive tumor microenvironment but also introduces LLPS suppression as a mechanism for TAM regulation, positioning SOP as a potent platform for enhanced chemo-immunotherapy.

KEYWORDS: liquid–liquid phase separation, tumor-associated macrophages, mTOR inhibition, carrier-free nanoassembly, chemo-immunotherapy



INTRODUCTION

Pancreatic ductal adenocarcinoma (PDAC), the predominant form of pancreatic cancer, remains one of the most lethal malignancies, with a five-year survival rate below 7%.^{1,2} By 2030, PDAC is projected to become the second leading cause of cancer-related death in the United States, surpassed only by lung cancer.³ These dire statistics highlight the urgent need for therapeutic approaches that can yield durable clinical benefit.

Cytotoxic chemotherapy remains central to PDAC treatment, not only because it induces tumor regression but also because it provokes immunogenic cell death capable of priming antitumor immunity.^{4,5} However, these benefits are severely undermined by the immunosuppressive tumor microenvironment, in which tumor-associated macrophages

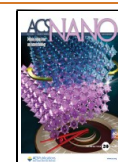
(TAMs) are key drivers of therapy resistance and disease progression.^{6,7} TAMs exist along a phenotypic continuum, with pro-inflammatory M1 macrophages promoting tumor clearance and immunosuppressive M2 macrophages sustaining tumor growth, angiogenesis, and immune evasion.^{8–10} Since the balance is tipped toward the M2 state, reprogramming

Received: October 3, 2025

Revised: July 2, 2026

Accepted: July 6, 2026

Published: July 13, 2026



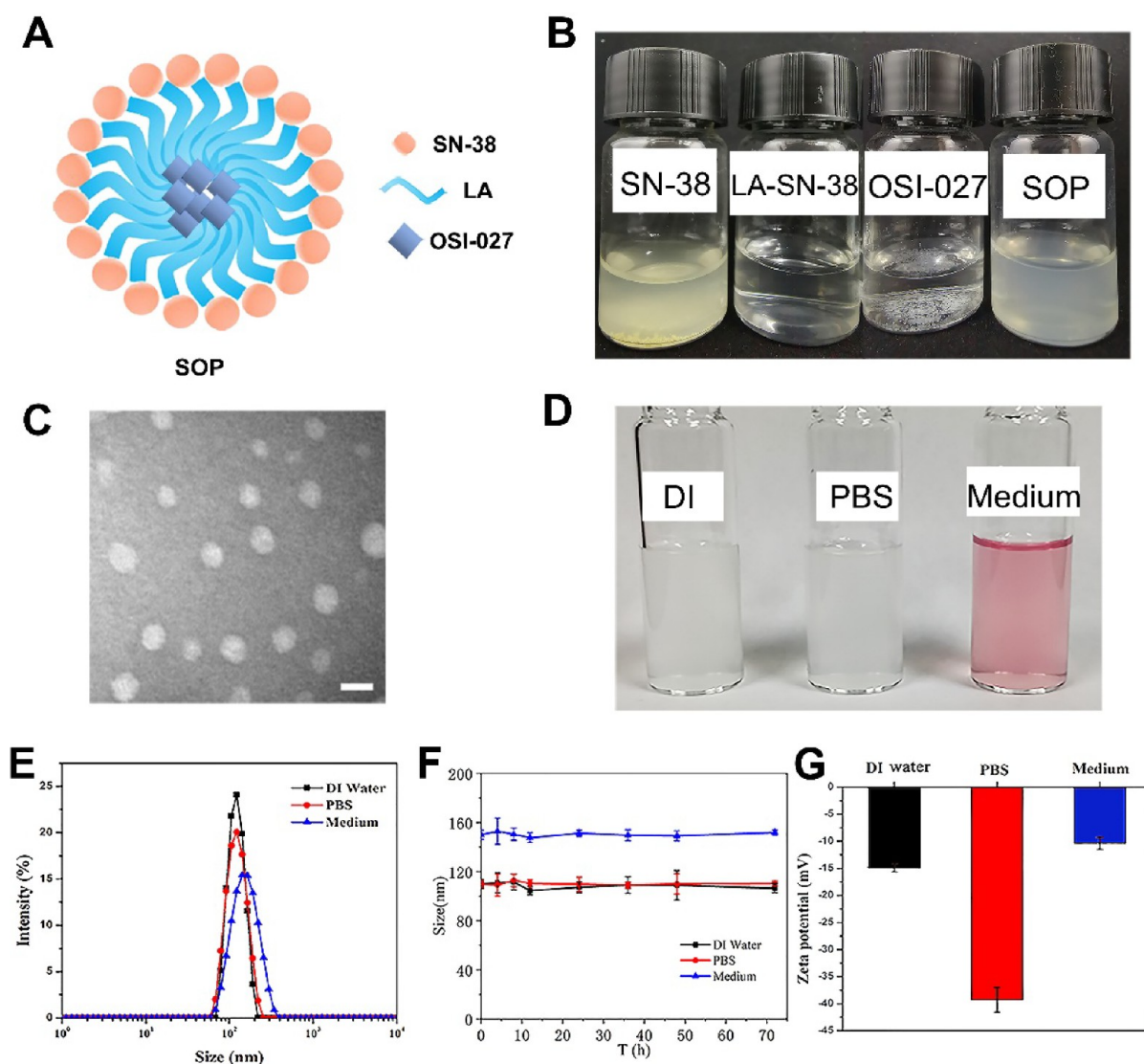


Figure 1. Characterization of SOP. (A) The structure of SOP. (B) LA-SN38 and SOP could form clear and transparent solutions in water, but free SN-38 and OSI-027 would precipitate. (C) TEM image of SOP. Scale bar: 100 nm. (D) Solubility, (E) particle-size distribution, (F) particle-size changes at different time points and (G) zeta potentials of SOP in DI water, PBS buffer and cell culture medium containing 10% FBS. Data are shown as mean $\pm$ SD ($n = 3$).

TAMs toward a pro-inflammatory M1 phenotype has thus emerged as a critical yet challenging therapeutic axis.

Conventional strategies for TAM repolarization primarily target cytokine and signaling pathways,¹¹ yet mounting evidence indicates that liquid–liquid phase separation (LLPS) also governs immune cell states by controlling the assembly of signaling and transcriptional condensates.^{12,13} Dysregulated LLPS sustains the M2 program, thereby reinforcing immunosuppression.¹⁴ Crucially, ribosomes have been identified as dominant molecular crowders that enhance LLPS, and their abundance is directly tuned by mammalian target of rapamycin complex 1 (mTORC1) signaling.¹⁵ This establishes a previously underappreciated biophysical layer of immune regulation: mTORC1-driven ribosome biogenesis creates a crowded cytoplasmic environment that facilitates phase separation, thereby stabilizing the M2 phenotype.¹⁵ Accordingly, disrupting this ribosome-LLPS axis offers a conceptual departure in TAM reprogramming.

While selective mTORC1 inhibitors show partial efficacy,¹⁶ they spare mTORC2, which limits their ability to fully suppress

ribosome biogenesis and TAM M2 polarization.^{17,18} In contrast, the dual mTORC1/2 inhibitor OSI-027 not only blocks protumorigenic cytokine programs but also directly reduces ribosome abundance, thereby attenuating ribosome-driven LLPS and shifting TAMs toward an M1-like state.^{19,20} However, the poor aqueous solubility of both OSI-027 and conventional chemotherapeutics poses a barrier to clinical translation, and nanocarriers based on exogenous materials raise concerns about off-target toxicity and metabolic degradation.^{21–24}

To address these challenges, we developed a carrier-free nanoassembly (SOP) in which 7-ethyl-10-hydroxycamptothecin (SN-38), conjugated to linoleic acid (LA), self-assembles and encapsulates OSI-027. This design enables efficient codelivery while harnessing a ribosome-inspired crowding control mechanism to suppress LLPS, a strategy that has not been exploited for TAM reprogramming. Upon intravenous administration, SOP preferentially accumulates in tumors, releases SN-38 to trigger immunogenic cell death, and delivers OSI-027 to inhibit mTORC1/2, reduce ribosome abundance,

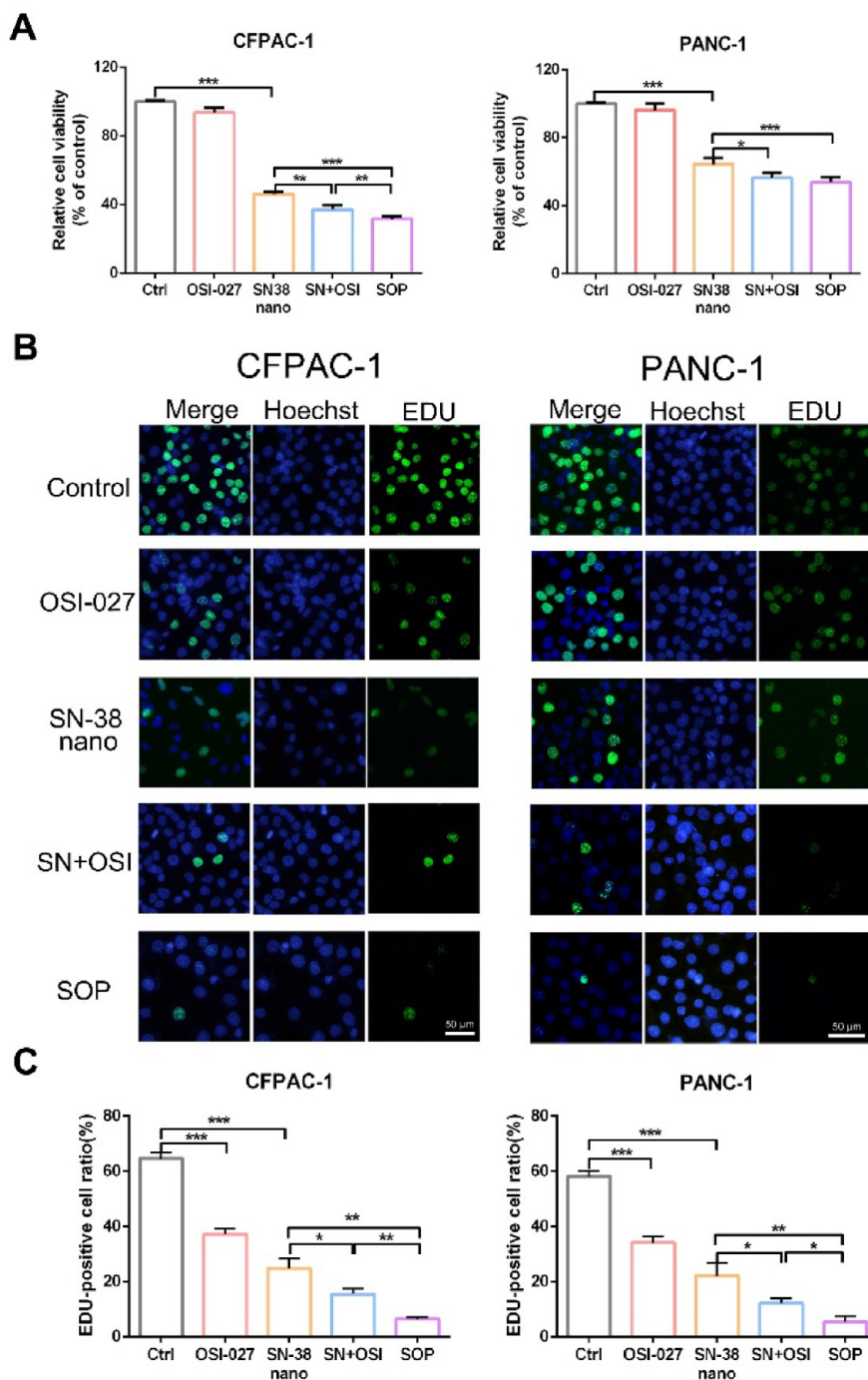


Figure 2. SOP suppresses PDAC cell growth and proliferation *in vitro*. (A) CFPAC-1 and PANC-1-cell viability after treatment with SOP (SN-38: OSI-027 = 2:1). (B) CFPAC-1 and PANC-1-cell proliferation with EdU labeling images and (C) quantifications of EdU-positive cells in percentages. Scale bar: 50 μ m. Data are shown as mean $\pm$ SD ($n = 3$). (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$).

suppress LLPS, and reprogram TAMs toward an M1 phenotype.

Through this dual molecular-biophysical mechanism, SOP establishes LLPS suppression as a mechanistic axis for chemo-immunotherapy and demonstrates potent efficacy across cell-derived xenograft, patient-derived xenograft, and genetically engineered KPC mouse models of PDAC. Collectively, these findings highlight the untapped potential of carrier-free nanoassembly that integrate drug delivery with active modulation of intracellular biophysics, advancing the frontier in TME remodeling and chemo-immunotherapy.

RESULTS

Preparation and Characterization of SOP

Due to the hydrophobic nature of OSI-027, nanoparticles with hydrophobic core structures were employed to improve its water solubility. Based on a previous study,²⁵ SN-38 modification was performed via a one-step esterification reaction: LA and SN-38 were reacted at a 1:1 molar ratio, and then EDC-HCl, DMAP and DIEA were added at a 1:1:1 molar ratio at 25 $^{\circ}$ C overnight. After removal of the solvent, dichloromethane was added, and the mixture was sequentially

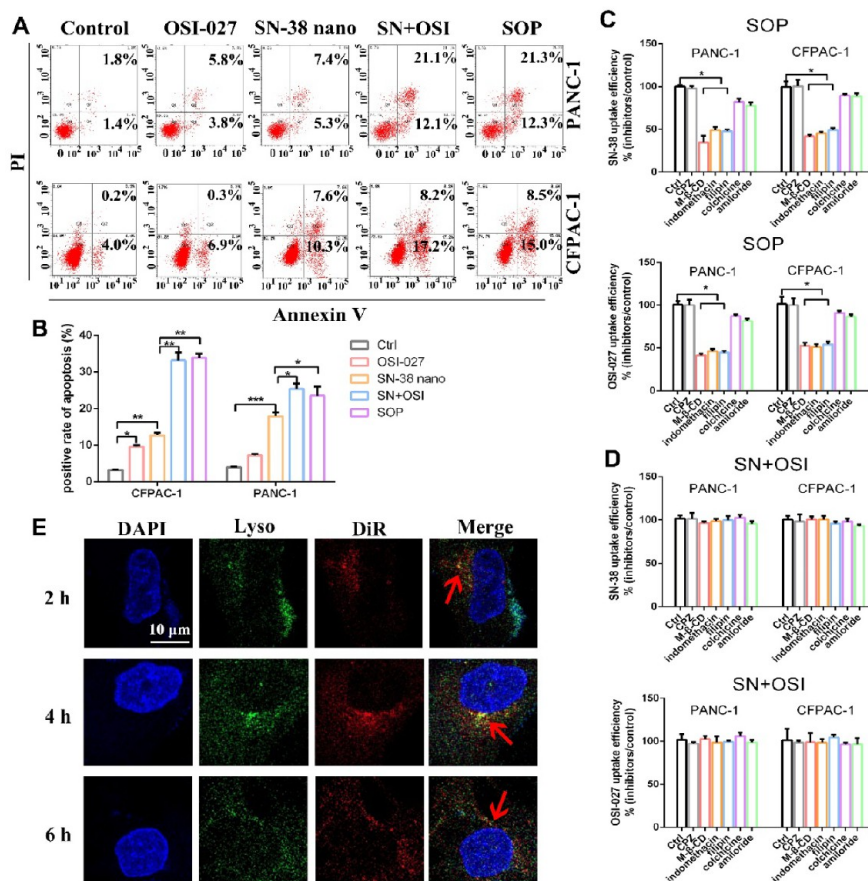


Figure 3. SOP induces PDAC cell apoptosis *in vitro*. (A) CFPAC-1 and PANC-1-cell apoptosis determined by an Alexa Fluor 488 Annexin V/PI staining kit and (B) quantification of the positive rate of apoptosis as a percentage. Effects of inhibitors on the cellular uptake of SN-38 and OSI-027 via SOP (C) and SN+OSI (D) quantified by HPLC. The cells without inhibitors were regarded as the control. (E) CLSM images of PANC-1 cells after treatment with SOP. LysoTracker, SOP and nuclei in green, red and blue, respectively. Scale bar: 10 μ m. Data are shown as mean $\pm$ SD ($n = 3$). (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$).

washed with 5% aqueous citric acid, saturated aqueous sodium bicarbonate, and brine. Then the organic layer was dried over anhydrous sodium sulfate, filtered, and concentrated under vacuum. The resulting residue was purified by flash column chromatography on silica gel (eluent: DCM/MeOH = 100:1, v/v), yielding LA-SN-38 with a yield of 65.8%. Its structure was confirmed by nuclear magnetic resonance (NMR, Figure S2) and ultrahigh performance liquid chromatography—triple quadrupole—time-of-flight mass spectrometry (UPLC–TripleTOF/MS, calculated value $[C_{40}H_{50}N_2O_6]^+ [M + H]^+ = 655.3742$; observed value 655.3737), and was consistent with the results reported.

LA-SN-38 was prepared to solve the aqueous solubility problem of SN-38 and OSI-027 and conveniently realize their codelivery. According to the previous study,²⁵ LA-SN-38 was first dissolved in dimethyl sulfoxide (DMSO), followed by injection into water for the self-assembly process to obtain SN-38 nanoparticles (SN-38 nano). The nanostructure was facilitated by the high structural flexibility of LA, and strong π - π stacking of SN-38. The nanostructure was facilitated by the high structural flexibility and possible intermolecular π - π stacking of LA, and strong π - π stacking between planar structures of SN-38.²⁵ OSI-027 was subsequently introduced into the core of SN-38 nano by hydrophobic interactions to finally obtain SOP (Figure 1A). The encapsulation efficiency (EE) and loading capacity (LC) were calculated as 98% and

18%, respectively, via high-performance liquid chromatography (HPLC, Figure S3).

SOPs with various ratios of SN-38 and OSI-027 were obtained by adding different quantities of OSI-027 to SN-38 nano. And the particle sizes and morphologies of different concentrations of SOP were shown in Figure S4. The nanoparticle size was critically dependent on the initial SOP concentration. Formulations at 0.5, 1, and 2 mg/mL consistently yielded uniform nanoparticles of approximately 110 nm. However, increasing the concentration to 4 mg/mL resulted in a sharp size increase to ~ 350 nm. This transition reflects a shift from a thermodynamically trapped, self-limiting growth regime at lower concentrations, where efficient steric stabilization by LA chains favors nucleation over growth, to a kinetically driven aggregation regime at higher concentrations.²⁶ At 4 mg/mL, excessive nucleation density and elevated local concentration of LA-SN-38 and OSI-027 overwhelm the steric stabilization capacity, inducing particle fusion to minimize surface energy.²⁷ Therefore, SOP concentrations ≤ 2 mg/mL were used in subsequent experiments.

The aqueous solubilities of SN-38, LA-SN-38, OSI-027 and SOP are shown in Figure 1B, where free SN-38 and OSI-027 had visible precipitates, while LA-SN-38 formed a transparent solution and SOP formed a homogeneous dispersion. When LA-SN-38 and SOP were stored at 4 $^{\circ}$ C for a week, no further precipitation was observed. Thus, the obtained SOP

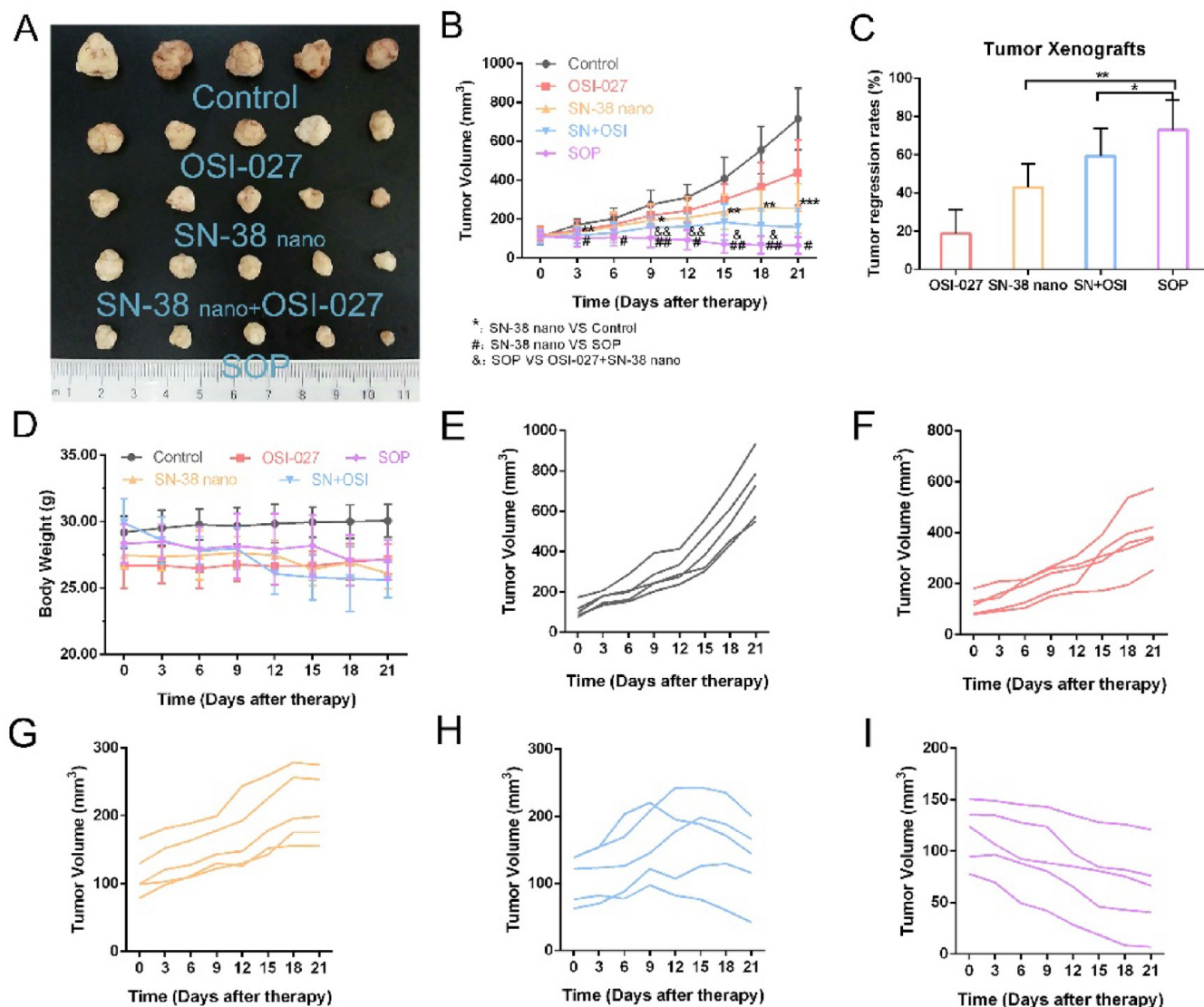


Figure 4. *In vivo* antitumor effect of SOP. (A) Tumor images of the control, OSI-027, SN-38 nano, SN+OSI and SOP groups. (B) Tumor volume curves, (C) tumor regression rates and (D) body weight change curves of mice in PDAC tumor xenografts. (E–I) Individual tumor volume curves of the control, OSI-027, SN-38 nano, SN+OSI and SOP groups. Data are shown as mean $\pm$ SD ($n = 5$). (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$).

successfully resolved the water insolubility of SN-38 and OSI-027, making its clinical application possible.

Given the success of SOP preparation, the morphology of SOP was characterized. As illustrated by transmission electron microscopy (TEM, Figure 1C), the typical nanostructure presented a narrow size distribution of 100–110 nm. SOP were well dispersed in deionized (DI) water and PBS buffer (Figure 1D) with an average diameter of 110 nm (Figure 1E), which was consistent with the TEM result. When SOP was added to cell culture medium containing 10% fetal bovine serum, its size slightly increased but remained up to 150 nm, as shown by dynamic light scattering analysis (DLS, Figure 1E). This observation may be attributed to the high ionic strength of the culture medium, which can perturb the hydration layer at the nanoparticle surface and consequently increase the hydrodynamic diameter.^{28,29} In addition, there were no observable diameter changes in the following 3 days (Figure 1F), indicating the stability of SOP in different media. SOP bore negative charges in all three media (Figure 1G), which

reduced the possibility of it binding to proteins and endovascular cells and prevented its removal from blood circulation.^{30,31} Therefore, SOP could be further applied in biological systems for clinical application.

In Vitro Cytotoxicity and Apoptosis Assays Induced by SOP and the Internalization Mechanism of SOP

The *in vitro* cytotoxicity of groups with various S/O ratios was investigated by CCK-8 assay. As shown in Figure S5A, free SN-38 showed a significant cytotoxic impact as expected, whereas negligible cytotoxicity was observed in free OSI-027. Interestingly, OSI-027 enhanced the ability of SN-38 to inhibit CFPAC-1 and PANC-1 cells. Especially at an S/O ratio of 2, OSI-027 lowered cell viability by 21.2% for CFPAC-1 cells and 8.1% for PANC-1 cells compared to free SN-38, resulting in the lowest half maximal inhibitory concentration (IC₅₀). Therefore, an S/O ratio of 2 was used for all further SOP preparations, and the IC₅₀ values of CFPAC-1 and PANC-1-cell proliferation were 0.5 μ M and 5 μ M, respectively (Figure S5B).

We next evaluated the cytotoxicity of OSI-027, SN-38 nano, SN+OSI (2:1) and SOP. CCK8 (Figure 2A) results showed that free OSI-027 did not significantly inhibit PDAC cell proliferation but could facilitate SN-38 inhibition of cell proliferation. EdU assays (Figure 2B–C) suggested that free OSI-027 not only significantly inhibit PDAC cell proliferation but also could facilitate SN-38 inhibition of cell proliferation. Thus, OSI-027 and SN-38 synergistically enhanced antiproliferative activities, which clearly suggested that drug combination therapy could successfully inhibit PDAC cell growth and proliferation. However, it is of note that the cytotoxic effects of SN+OSI and SOP *in vitro* were nearly equivalent. We attributed this finding to the fact that SN-38 nano- and liposoluble OSI-027 both readily entered cells in the above experiments *in vitro*, resulting in the integration of OSI-027 with SN-38 having little distinction from SOP on cell proliferation *in vitro*.

This phenomenon also occurred in the apoptosis assay (Figure 3A–B). SN-38 induced apoptosis in PDAC cells, while OSI-027 did not significantly induce PANC-1 cell apoptosis, but the ability of SN+OSI to induce apoptosis was enhanced, which was consistent with SOP. In conclusion, there was almost no difference in the performance of PDAC cell-based experiments *in vitro* between SN+OSI and SOP. Therefore, it was necessary to study the drug uptake behavior of PDAC cells in response to SOP and SN+OSI. Different inhibitors were used to investigate the route of drug entry into cells. HPLC was employed to quantify the concentrations of OSI-027 and SN-38. Importantly, regardless of whether the detected SN-38 was in its free form or conjugated as LA-SN-38, its presence inside cells served as direct evidence that the nanoparticle formulation was successfully internalized. The collected data (Figure 3C) demonstrated that chlorpromazine (CPZ, clathrin-mediated uptake inhibitor), colchicine and amiloride (macropinocytosis inhibitors) had no effect on SOP uptake by PDAC cells. 2,6-dimethyl- β -cyclodextrin (M- β -CD), indomethacin and filipin (caveolin-mediated uptake inhibitors) significantly decreased the cellular uptake of SOP, since the accumulation of SN-38 and OSI-027 in the cells of these groups was less than that in the control. On the other hand, for the free SN+OSI groups (Figure 3D), none of the inhibitors had an effect. In conclusion, these results indicated that drug uptake at the IC₅₀ was efficiently achieved by PDAC cells via free diffusion in the free SN+OSI groups and via the caveolin-mediated endocytic pathway in the SOP groups *in vitro*. In this way, for SOP and SN+OSI at IC₅₀, the drug accumulation in PDAC cells was nearly equivalent, leading to similar effects on cell growth, proliferation and apoptosis induction *in vitro*. As illustrated in Figure 4E, confocal microscopy revealed that following cellular internalization, the labeled SOP initially underwent progressive lysosomal trafficking, reaching peak accumulation within lysosomal compartments at approximately 4 h postinternalization.^{32,33} Subsequently, they exhibited efficient lysosomal escape.^{32,33} Upon exposure to the acidic lysosomal milieu (pH $\sim$ 5.0), the ester bonds within the LA-SN-38 underwent hydrolysis, thereby triggering a controlled and sustained drug release.³⁴

In Vivo Antitumor Efficacy of SOP

To assess the performance *in vivo* and explore the potential of further clinical translation, experiments of antitumor efficacy were carried out sequentially. A cell-derived xenografts model (CDX, Figure S6) was preliminarily adopted and revealed that

combined pharmacotherapy was far more effective than separate ones. In addition, SOP, with equal drug content, exhibited higher inhibition of tumor growth than SN+OSI. SOP does not cause significant damage to the liver or kidney compared to SN-38 and SN+OSI, which reduces the cytotoxicity of drugs (Figure S7A–D). The histological examination via hematoxylin and eosin (H&E) staining of major organs, including the heart, liver, spleen, lungs, and kidneys, further corroborated the favorable *in vivo* biosafety profile of the SOP (Figure S7E). Notably, encapsulation of OSI-027 within the SOP markedly attenuated its systemic toxicity, as evidenced by a significant reduction in drug-induced cardiotoxicity and nephrotoxicity compared to administration of the free drug. The antitumor efficacy of SOP was also confirmed by positron emission tomography (PET) images of mice in the KPC model (Figure S8). The patient-derived xenografts (PDX) model was subsequently applied to simulate the tumor microenvironment and verify the above results. As shown in Figure 4A, compared to the saline control, tumor growth was obviously restrained after tail intravenous injection of OSI-027, SN-38 nano, SN+OSI and SOP, resulting in average tumor volumes of 438.6 ± 168.0 mm³, 255.1 ± 127.1 mm³, 159.5 ± 96.3 mm³ and 64.4 ± 42.5 mm³, respectively, as opposed to 715.3 ± 158.7 mm³ of the saline control (Figure 4B). Correspondingly, the tumor regression rates (Figure 4C) were 19.8%, 42.5%, 59.9% and 73.1%, respectively, corroborating that SOP had the greatest tumor inhibition *in vivo*. Notably, the tumor volume of the group treated with SOP decreased steadily during the end of the therapy period and even exhibited a tendency toward tumor extinction. In particular, the mouse weight (Figure 4D) of the SN+OSI group decreased significantly, while the other groups remained unchanged in general. These results might contribute to the enhanced permeability and retention effect (EPR) of nanoparticles.³⁵ Compared to direct drug combination therapy, it was easier for the nanoparticles made up of SN-38 and OSI-027 to penetrate into and stay in the tumor tissue. Furthermore, the SOP formulation exhibited a favorable pharmacokinetic profile. The elimination half-lives of SN-38 and OSI-027 were both approximately 8 h, as determined from the plasma concentration–time curves (Figure S9). This enhanced intratumoral accumulation was further corroborated by biodistribution of SOP *via in vivo* imaging system and fluorescence staining of tumor sections. SOP were labeled with the near-infrared fluorescent dye DiR and administered intravenously, after which whole-body fluorescence signals were monitored over time. As shown in Figure S10A, mice injected with free DiR exhibited rapid systemic distribution of the dye within hours. In contrast, SOP gradually accumulated at the tumor site, remaining detectable for up to 24 h. Ex vivo fluorescence imaging was also exhibited the substantial higher fluorescence intensity in tumor tissue of the SOP group (Figure S10B). Moreover, at 24 h postadministration of DiR-labeled SOP, pronounced extravasation from CD31-labeled blood vessels into the tumor parenchyma was observed, particularly within perinecrotic and stromal regions (Figure S10D). Therefore, the therapeutic effect on tumor tissues and the biosecurity on normal tissues were improved in the meantime.

In Vivo Cell Proliferation of SOP

The antitumor efficiency *in vivo* was then investigated by more detailed analysis. The proliferation of tumor cells *in vivo* was

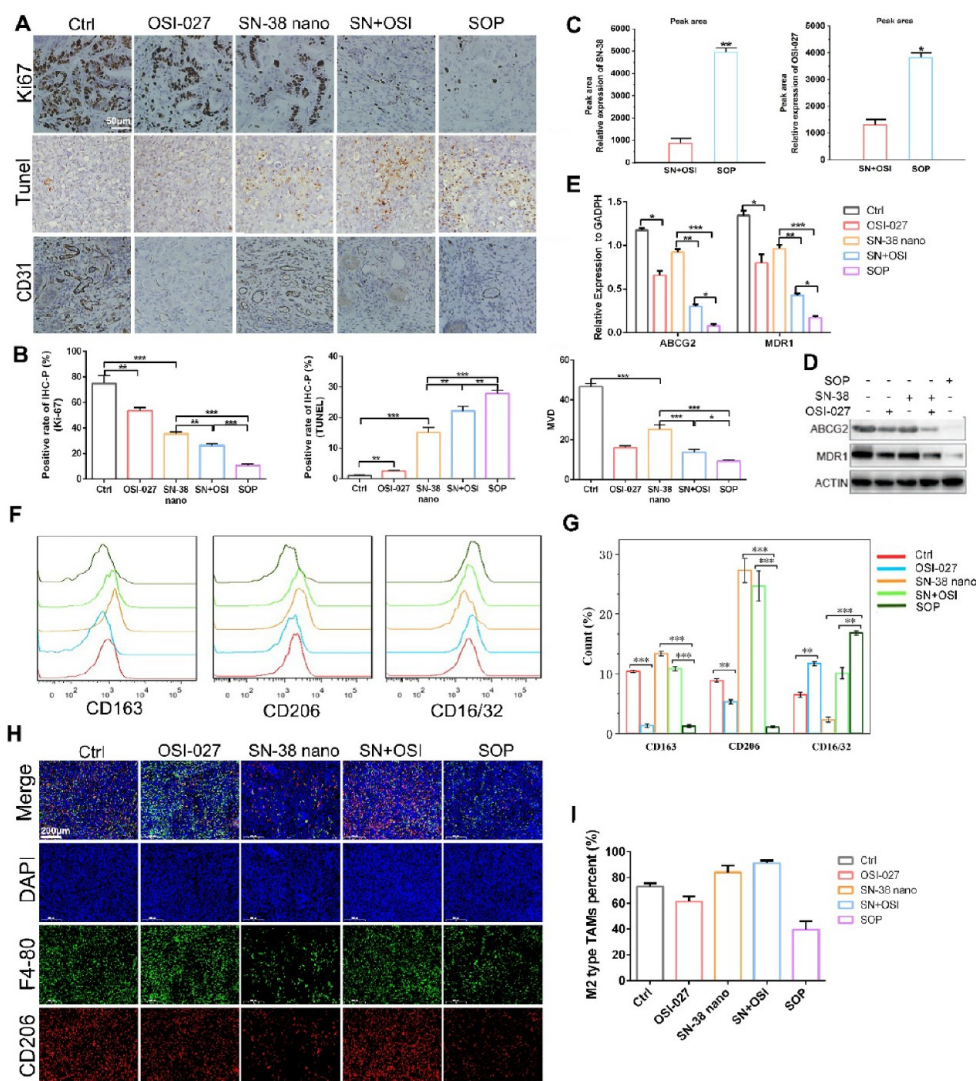


Figure 5. Potential mechanism of SOP's effects *in vivo* on PDAC tumor models. (A) *Ki-67*-stained cellular proliferation, TUNEL-stained pathological changes, and CD31 antibody-stained microvasculature in tumor tissues and (B) quantifications of microvascular density in percentages. (C) The concentrations of SN-38 and OSI-027 in the tumor site via HPLC. (D) Expression of ABCG2 and MDR1 in tumors after treatment and (E) quantification of gray values in percentages. (F) Expression of CD163, CD206 and CD16/32 in PBMCs via FACS and (G) quantification of percentages. (H) Fluorescence staining image of macrophage infiltration patterns and subtype distribution within the tumor microenvironment, and (I) quantitative evaluation of the M2 macrophage population proportion. Scale bar: 200 μm . Data are shown as mean $\pm$ SD ($n = 3$). (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$).

evaluated via immunohistochemical (IHC) staining of *Ki-67* (Figure 5A, first row). Cells were massively stained with *Ki-67* in the control group. After treatment with OSI-027 or SN-38 nano, the number of *Ki-67*-positive cells decreased slightly. In addition, the number further decreased with the combination therapy of SN+OSI and reached the lowest with the administration of SOP (Figure 5B). The results again validated that SOP remarkably inhibited cell proliferation in PDAC tumors and that SN-38 played an important role in this process.

In Vivo Cell Apoptosis Induced by SOP

TUNEL staining (Figure 5A, second row) was utilized to determine whether SOP induced apoptosis in PDAC tumor cells. There was minimal apoptosis ($1.09 \pm 0.15\%$) in the control group. In contrast, SOP ($27.82 \pm 0.67\%$) and SN+OSI ($22.13 \pm 0.85\%$) treatment significantly induced apoptosis of PDAC cells, while $15.13 \pm 0.96\%$ of apoptotic cells were

detected in the SN-38 nanotreated groups. In particular, the OSI-027 group ($2.44 \pm 0.16\%$) barely displayed apoptosis (Figure 5B), which means that OSI-027 does not have the capability to induce apoptosis but does promote it induced by SN-38, and SOP induced more cell apoptosis than SN+OSI.

In Vivo Antiangiogenic Activity by SOP

Since OSI-027 could remarkably suppress mTOR signaling, which is closely related to angiogenesis, we assessed the influence of SOP on angiogenesis. IHC staining was performed using a CD31 antibody (Figure 5A, third row), which is a direct marker of angiogenesis. Compared to the control group, the microvessel density (MVD) decreased in the SN-38 nano group and was much weaker after treatment with OSI-027 (Figure 5B). Furthermore, SOP exhibited higher potency in the inhibition of angiogenesis than the direct combination of SN+OSI, indicating that SOP greatly suppressed the angiogenesis of PDAC tumors.

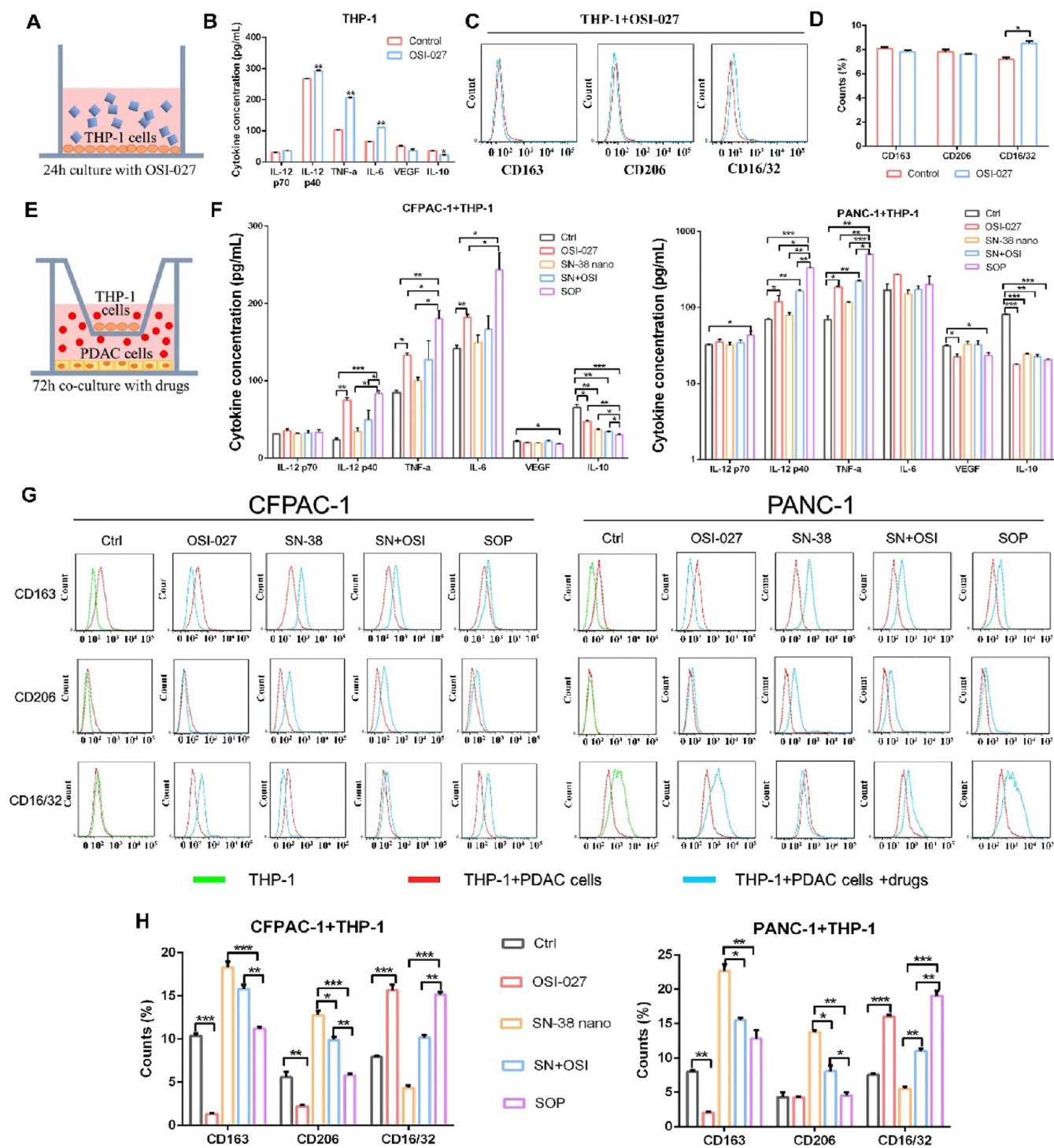


Figure 6. Effect of OSI-027 on THP-1 cells *in vitro*. (A) A schematic of the experiment with THP-1 cells treated with OSI-027. (B) Cytokine expression of THP-1 cells with or without OSI-027 treatment. (C) Expressions of CD163, CD206 and CD16/32 in THP-1 cells after treatment with OSI-027 via FACS and (D) Quantifications in percentages. (E) A schematic of the THP-1 and PDAC cell coculture system. (F) Cytokine expression in the THP-1 and PDAC cell coculture system after treatment. (G) Expression of CD163, CD206 and CD16/32 in the THP-1 and PDAC cell coculture system and (H) quantification of the percentages. Data are shown as mean $\pm$ SD ($n = 3$). (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$).

Quantification of Drug in Tumor Tissues

Since the therapeutic effect of SOP was better than SN+OSI when both drugs were administered at the same dose, we consequently analyzed the remaining drug level in tumors via HPLC. As shown in Figure 5C, the concentrations of SN-38 and OSI-027 in SOP tumors were both much higher than

those in the SN+OSI group, suggesting that more SN-38 and OSI-027 were delivered and stayed in SOP tumors. Herein, SOP, formed by encapsulating the mTOR inhibitor OSI-027 with the anticancer drug SN-38, not only improved the solubility of drugs but also delivered more of them to target positions for better treatment.

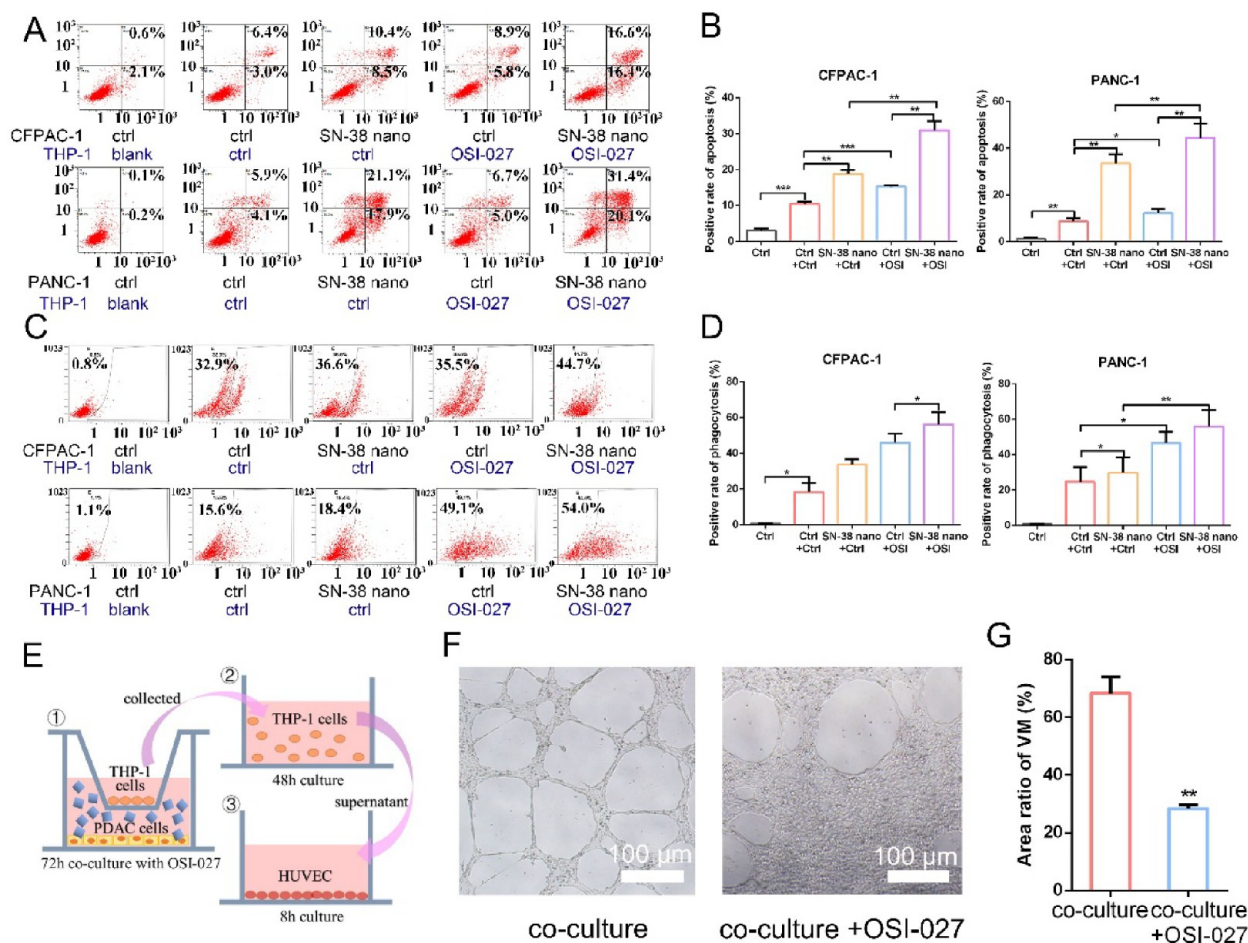


Figure 7. Mechanism of tumor growth inhibition by OSI-027. (A) Apoptosis of PDAC cells by the OSI-027-treated THP-1-cell system and (B) quantification of the positive rate of apoptosis as a percentage. (C) Phagocytosis of PDAC cells by the OSI-027-treated THP-1-cell system and (D) quantification of the positive rate of phagocytosis as a percentage. (E) A schematic of the HUVEC culture supernatant of the OSI-027-treated coculture system. (F) Microvascular density of the PDAC cell and THP-1-cell coculture system after OSI-027 treatment and (G) quantification of the area ratio of the microvascular density percentages. Data are shown as mean $\pm$ SD ($n = 3$). (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$).

Gene Expression Induced by SOP

In our previous study, OSI-027 was found to largely inhibit the expression of resistance genes.³⁶ Moreover, drug resistance is common in PDAC therapy,³⁷ and we subsequently evaluated the expression of resistance proteins isolated from tumor cells. The images (Figure 5D) revealed that MDR1 protein was downregulated after OSI-027 treatment, which was in accordance with our previous results. ABCG2, a resistance-related translocator, was also suppressed by OSI-027 treatment. As expected, OSI-027 significantly enhanced the inhibition of MDR1 and ABCG2 expression, which was conducive to the efficacy of SN-38, especially in SOP treatment *in vivo* (Figure 5E) and *in vitro* (Figure S11).

TAM Repolarization by SOP

As mentioned, mTOR signaling has been reported to play a vital role in regulating the phenotype of TAMs.⁸ TAMs are abundant in the tumor microenvironment and can differentiate into antitumor M1 Φ s and protumor M2 Φ s.^{8,10,38,39} M1 Φ promotes inflammatory and immune activities to suppress tumor growth.^{39,40} M2 Φ causes immunosuppression and promotes vascularization, thereby accelerating tumor progression.³⁹ There is evidence indicating that TAMs can be reprogrammed from the polarized M2 to the M1 phenotype to

alter the immunoregulatory response and affect the TME.^{39,41} Therefore, repolarization from the M2 to the M1 phenotype is a potential method in the treatment of tumors. In this study, we found that the expression of CD163 and CD206 in peripheral blood mononuclear cells (PBMCs, Figure 5F–G) and tumor tissues (Figure S12), as M2 markers, were decreased by OSI-027, SN+OSI and SOP in the TME, while both were enhanced by SN-38, indicating the ability of OSI-027 to suppress the M2 phenotype. Meanwhile, the M1 phenotype was increased in the TME of OSI-027, SN+OSI and SOP, as reflected by the upregulation of the M1 marker CD16/32. In other words, TAM repolarization occurred in the TME of SOP. The SN-38-induced M2 phenotype could reverse to the M1 phenotype under the impact of OSI-027.

To evaluate the impact of treatment on macrophage polarization within the tumor microenvironment, tumor sections from each group were subjected to immunofluorescence costaining for F4–80 (a pan-macrophage marker) and CD206 (a marker associated with the M2 phenotype). Quantitative analysis demonstrated that the SOP group exhibited the lowest overall abundance of TAMs among all treatment groups (Figure 5H–I). Furthermore, the proportion of M2-type TAMs (CD206⁺/F4–80⁺) was also significantly

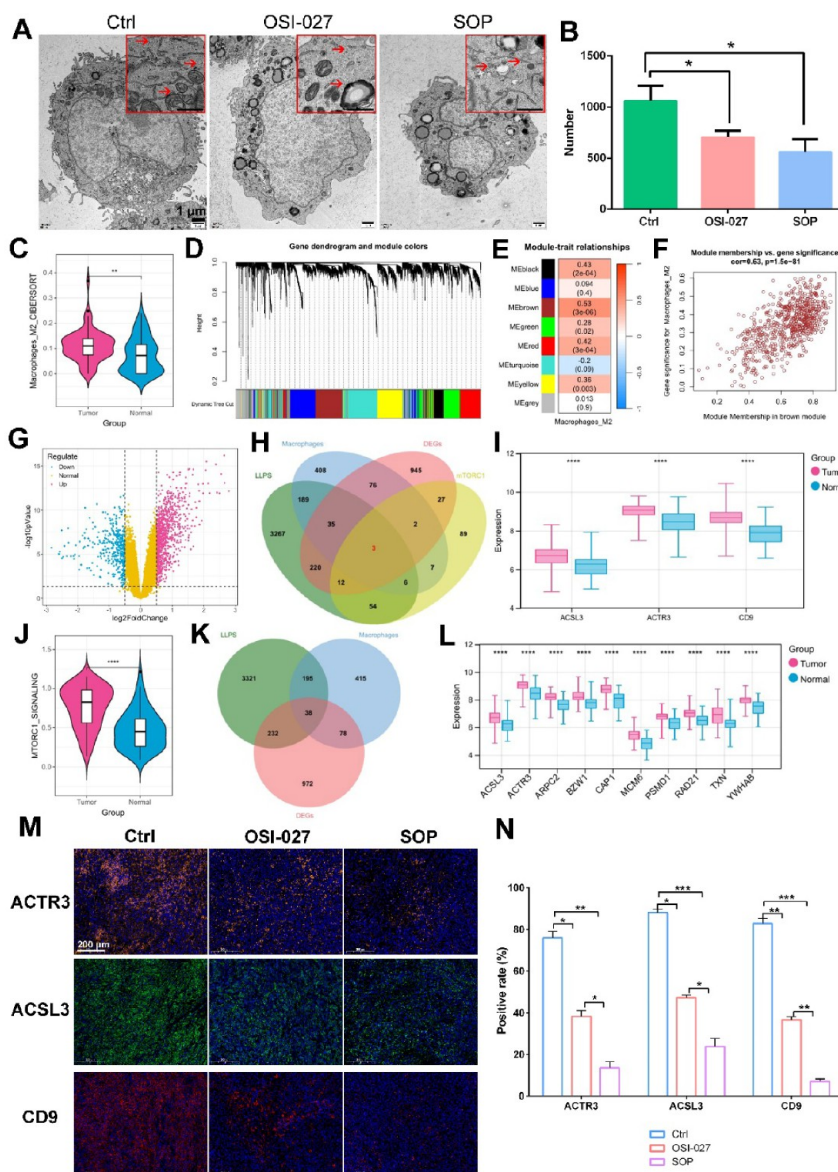


Figure 8. Mechanism of LLPS inhibition by OSI-27. (A) TEM images of THP-1 cells treated with OSI-27 and SOP and (B) its quantitative statistics of ribosomes. Scale bar: 1 μm . (C) Violin diagram of differential distribution of macrophage M2. (D) Results of gene clustering and module division fusion. (E) Module association analysis with phenotypic traits. Red indicates positive correlation, blue indicates negative correlation, and darker colors indicate stronger correlations. (F) Correlation between module membership (x-axis) and gene significance (y-axis). (G) Volcano map of DEGs. Blue represents downregulated genes and red represents upregulated genes. (H) Intersection Venn diagram of macrophage M2-related genes, DEGs, LLPS and mTORC1 and (I) resulted gene expression box plot. (J) mTORC1 signal distribution violin diagram. (K) Intersection Venn diagram of macrophage M2-related genes, DEGs and LLPS and (L) resulted gene expression box plot. (M) Tumor tissue fluorescence staining images of ACTR3, ACSL3 and CD9, and (N) its positive rate analysis. Scale bar: 200 μm . Data are shown as mean $\pm$ SD ($n = 3$). (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$).

reduced in this group, indicating a shift away from the immunosuppressive M2 phenotype. Collectively, these results provide evidence supporting the reeducation effect of OSI-27, which facilitates the repolarization of M2-polarized TAMs toward a less protumorigenic state.

Beyond the local immunomodulatory effects within the tumor microenvironment, we next investigated whether the SOP formulation could alleviate the systemic immunosuppression typically associated with free OSI-27 administration. To this end, serum cytokine profiles were evaluated in mice following various treatments. As illustrated in Figure S13, administration of free OSI-27 at 10 mg/kg induced a pronounced immunosuppressive state, characterized by sig-

nificantly reduced levels of the pro-inflammatory cytokines IL-2, TNF- α , and IL-6, alongside a marked elevation of the anti-inflammatory cytokine IL-10. A reduced dose of free OSI-27 (5 mg/kg) partially alleviated this immunosuppressive phenotype, yet systemic immunosuppression persisted. In striking contrast, encapsulation of OSI-27 (10 mg/kg) within the SOP formulation not only abrogated the immunosuppressive effects but also promoted an overall state of immune activation, as evidenced by significantly increased levels of pro-inflammatory cytokines (IL-2, TNF- α , IL-6, and IL-12) and decreased IL-10 expression. Collectively, these results demonstrate that the SOP formulation substantially broadens the therapeutic window of OSI-27 by uncoupling its immuno-

modulatory activity from the systemic immunosuppression otherwise imposed by the free drug.

The Effect of OSI-027 on the Phenotypic Polarization of Macrophages *In Vitro*

OSI-027 was used to culture human myeloid leukemia mononuclear cells (THP-1) to ascertain its impact on the phenotypic repolarization of TAMs. IL-12p70, IL-12p40, TNF- α and IL-6, antitumor cytokines, were all upregulated by OSI-027 treatment, whereas the protumor VEGF and IL-10 were downregulated (Figure 6B). Fluorescence-activated cell sorting (FACS) assays also showed that OSI-027 not only reduced the expression of CD163 and CD206 but also facilitated the expression of CD16/32 (Figure 6C–D). For the coculture system of PDAC and THP-1 cells, SOP treatment reversed M2 to M1 polarization by introducing OSI-027, as evidenced by the upregulation of CD16/32 and downregulation of CD163 and CD206 (Figure 6G–H), which was in sharp contrast to SN-38 treatment. This was further confirmed by the increased expression of IL-12p70, IL-12p40, TNF- α and IL-6 and the decreased expression of IL-10 (Figure 6F), which was consistent with the reported effect of mTOR inhibition on macrophage polarization.

OSI-027-Induced Macrophage Polarization Inhibited Tumor Growth by Promoting SN-38-Induced Cell Death, Engulfing Cancer Cells, and Suppressing Angiogenesis

We further detected how OSI-027-induced macrophage polarization inhibited tumor growth. First, cytokines secreted by THP-1 cells after treatment with OSI-027 could trigger apoptosis of PDAC cells, regardless of whether they were treated with SN-38 (Figure 7A–B). The results suggested that OSI-027 gave rise to PDAC cell apoptosis *in vivo* in a macrophage-related manner. Second, THP-1 cells treated with OSI-027 were more likely to engulf SN-38-pretreated PDAC cells, which served as M1 Φ cells (Figure 7C–D). Since angiogenesis is an essential step in tumor growth and VEGF has been reported to be blocked by inhibiting the mTOR signaling pathway, we evaluated the impact of OSI-027 in regulating angiogenesis via HUVECs (Figure 7E). After growing on Matrigel for 8 h, HUVECs without treatment formed capillary structures, while tube formation was suppressed in the presence of OSI-027 (Figure 7F–G). M2 Φ induced tube formation, but M1 Φ inhibited it; therefore, the induction direction was reversed by OSI-027 through M2-to-M1 repolarization.

The Intersection Analysis of LLPS and mTOR

As illustrated, treatment with OSI-027 and SOP significantly reduced ribosome abundance in both human THP-1 monocytic cells (Figure 8A–B) and murine RAW264.7 macrophages (Figure S14). Ribosomes, as a type of macromolecular crowding agent in the cytoplasm, are characterized by their large volume and high abundance, occupying significant space within the cellular environment.^{15,42} Research indicates that the physical crowding induced by ribosomes enhances depletion attraction effects; specifically, macromolecules tend to aggregate due to volume exclusion effects, thereby facilitating phase separation.¹⁵ Consequently, a reduction in ribosome concentration is likely to diminish instances of LLPS. Herein, this observation may be attributed to OSI-027-induced inhibition of mTORC1 activity, which subsequently reduces ribosome synthesis and lowers ribosomal concentrations, thereby attenuating LLPS.

Based on the expression profile data, the level of macrophage M2 was calculated, which showed significantly higher in the tumor than normal (Figure 8C). Then, The Weighted Gene Coexpression Network Analysis (WGCNA) was performed using macrophage M2 as the phenotypic trait, and hierarchical clustering was performed on the data (Figure S15A). And the soft threshold necessary for constructing the network was determined. We selected the soft threshold at which R^2 first exceeded 0.85, resulting in a value of 8. Under this soft threshold, the network approximates a scale-free distribution, and the mean number of gene connections gradually stabilizes, aligning with established principles (Figure S15B). Based on this soft threshold, an adjacency matrix was constructed; subsequently, we calculated the gene topology matrix and estimated dissimilarity coefficients between genes to obtain a systematic clustering tree among them. As shown in Figure 8D, module division was performed using a hybrid dynamic tree cutting algorithm, followed by merging similar modules. The minimum number of genes required for each module was set to 200, ultimately yielding a total of eight distinct modules. After module division, we assessed the correlation between each module and macrophage M2 (Figure 8E). Notably, the brown module ($n = 726$; $r = 0.53$; $p\text{-value} = 3 \times 10^{-6}$) exhibited the strongest significant correlation (Figure 8F).

According to the described methodology, differential expression analysis between tumor and normal tissues was performed. And a total of 1,320 differentially expressed genes were identified, comprising 937 upregulated genes and 383 downregulated genes (Figure 8G). As shown in Figure 8H, three intersecting genes (ACSL3, ACTR3 and CD9) were identified by cross-linking macrophage M2-related genes (WGCNA module genes), DEGs, LLPS and mTORC1, and it was found that all three genes were significantly overexpressed in the tumor group (Figure 8I).

The GSVA algorithm was employed to quantify mTORC1 signaling levels, with Figure 8J demonstrating that tumor-associated mTORC1 signaling levels were significantly higher than normal tissues. Subsequently, cross-identification of 38 genes through macrophage M2-associated gene sets (WGCNA module genes), DEGs, and LLPS revealed 10 genes showing significant correlations with mTORC1 signaling levels (Figure 8K). As shown in Figure 8L, using a threshold of $\text{cor} > 0.7$ and $p < 0.001$, these 10 genes were identified as significantly upregulated in tumor tissues compared to normal counterparts.

To validate this hypothesis, fluorescence immunostaining was performed on tumor sections treated with OSI-027 or SOP to assess the expression levels of three intersection genes (ACTR3, ACSL3, and CD9). As shown in Figure 8M–N, upon treatment, the number of condensates mediated by these three phase-separation genes was significantly reduced, providing direct visual evidence supporting the inhibition of LLPS. Notably, the reduction was markedly more pronounced in the SOP group, consistent with the enhanced delivery and bioavailability of OSI-027 conferred by the SOP formulation. These findings collectively indicate that OSI-027 effectively inhibits mTORC1 activity, promotes repolarization of M2-like tumor-associated macrophages, and attenuates LLPS processes via ribosomes.

DISCUSSION

One of the major barriers to effective PDAC therapy is the inability of drugs to penetrate the desmoplastic and poorly vascularized tumor stroma.^{37,43} Compounding this challenge,

PDAC is characterized by dense inflammatory infiltration where malignant epithelial cells comprise only a minority of the tumor mass.⁴⁴ Within this milieu, TAMs emerge as dominant regulators of therapeutic response, and their polarization state critically influences the efficacy of chemotherapy.⁴⁴ In particular, mTOR signaling has been identified as a central axis that integrates extracellular and intracellular cues to shape TAM phenotype, thereby governing the TME.⁴⁵ Conventional views hold that mTOR inhibition reprograms TAMs primarily by modulating cytokine and growth factor expression; additionally, suppression of mTOR signaling decreases VEGF and HIF-1 α levels, thereby attenuating angiogenesis and restraining tumor progression.^{45,46}

Our study extends this paradigm by revealing an additional, biophysical layer of regulation. Recent work has shown that mTORC1 controls ribosome abundance, and ribosomes act as dominant crowding agents that promote LLPS within the cytoplasm.¹⁵ While existing strategies for macrophage repolarization have largely converged on a defined set of molecular targets,^{47,48} most notably the Syk and STAT signaling pathways, the IL-4 receptor axis, and metabolic reprogramming of cellular bioenergetics, the potential link between ribosome-inspired macromolecular crowding, intracellular LLPS behavior, and macrophage polarization has, to our knowledge, not been previously articulated. The principal conceptual advance of the present study is therefore to propose a previously unrecognized regulatory axis, namely that ribosome-associated crowding can serve as a modulator of LLPS and, by extension, a contributor to the M2-to-M1 repolarization program of TAMs. We anticipate that this perspective will provide a useful starting point for future investigations into the broader landscape of polarization-regulatory mechanisms and will inform the design of next-generation immunomodulatory strategies.

LLPS stabilizes the transcriptional and signaling condensates that reinforce the immunosuppressive M2 state of TAMs. By achieving dual inhibition of mTORC1/2, OSI-027 not only blocked protumor cytokine programs but also reduced ribosome-driven crowding, thereby suppressing LLPS and shifting TAMs toward the pro-inflammatory M1 phenotype. This dual molecular–biophysical regulation provides a mechanistic rationale for the potent reeducation of TAMs observed in our PDAC models, reflected by increased secretion of IL-12, TNF- α , and IL-6 and decreased expression of IL-10 and VEGF. Notably, OSI-027 exhibited only minimal direct cytotoxicity against tumor cells *in vitro*, suggesting that its primary target is not the tumor cells themselves. Conversely, its marked therapeutic efficacy *in vivo* points to an immunomodulatory mechanism of action. This is consistent with our mechanistic findings: OSI-027, delivered via SOP nanoassembly, effectively reduces ribosome biogenesis in macrophages, thereby alleviating crowding-induced LLPS and promoting M1 polarization. Thus, its *in vivo* antitumor effect is primarily attributed to reprogramming the tumor immune microenvironment through macrophage repolarization, rather than direct tumor cell killing.

The carrier-free SOP nanoassembly proved crucial in realizing these effects. Encapsulation of OSI-027 within LA-SN-38 assemblies enabled preferential accumulation in tumors while minimizing systemic exposure, thereby reducing the risk of mTOR inhibitor-induced systemic immunosuppression. Upon release, OSI-027 reached sufficient concentrations to inhibit mTORC1/2, remodel ribosome abundance, and

disrupt LLPS. Concurrently, SN-38, a topoisomerase I inhibitor, blocked DNA replication and RNA synthesis, producing immunogenic tumor cell death that synergized with TAM reprogramming.⁴⁹ Importantly, OSI-027 also downregulated MDR1 expression, enhancing PDAC cells sensitivity to SN-38. Together, these mechanisms produced a reinforced therapeutic loop: SN-38 killed tumor cells and activated antitumor immunity, while OSI-027 both reprogrammed macrophages and alleviated drug resistance, resulting in synergistic tumor suppression.

These findings highlight the importance of targeting the TME through both molecular and biophysical mechanisms. Although the cellular interactions within the TME remain highly complex, our data suggest that modulation of ribosome-driven LLPS represents an underexplored lever for therapy. By uniting TAM reprogramming, angiogenesis suppression, drug-resistance reversal, and immunogenic cell death, SOP offers a multifaceted approach to remodel the PDAC microenvironment. More broadly, this work demonstrates that carrier-free nanoassemblies can serve not only as delivery vehicles but also as active modulators of intracellular biophysics, establishing a distinct paradigm for the design of chemo-immunotherapeutic platforms. In the future, further investigation into key immunomodulatory mechanisms, including immunogenic cell death, dendritic cell activation, and T cell recruitment, is essential to elucidate the comprehensive role of this engineered system in immune regulation.

Study Limitations and Future Directions

It is important to acknowledge several limitations of the present study. First, while our data establish the plausibility of the proposed ribosome–crowding–LLPS–polarization axis, the mechanistic dissection of a single LLPS-driving protein is, in itself, often sufficient to support an independent publication, and such a comprehensive dissection lies beyond the scope of the present manuscript. The current work is therefore intended as a foundational report that establishes the tripartite association, rather than a complete mechanistic proof. Second, although our bioinformatic integration and immunofluorescence analyses have converged on a panel of candidate LLPS proteins, additional experimental validation is required to definitively establish which of these candidates is the principal driver of the observed phenotype. We wish to ensure the reader that, building on the present findings, we are actively pursuing a series of in-depth follow-up investigations, including targeted gene knockout and overexpression, protein domain truncation, and site-directed mutagenesis of the candidate proteins, with the aim of delineating the precise molecular determinants of the proposed ribosome-crowding-LLPS-polarization axis. We consider these studies to be of sufficient scope and depth to warrant a separate, dedicated investigation, and we hope that the present manuscript, together with the proposed follow-up work, will provide a useful reference for other investigators working at the interface of LLPS biology and tumor immunology.

CONCLUSION

In summary, we present a carrier-free nanoassembly (SOP) that codelivers the dual mTORC1/2 inhibitor OSI-027 and the chemotherapeutic agent SN-38 to remodel the PDAC microenvironment through a dual mechanism integrating molecular and biophysical control. OSI-027 reduces ribosome abundance, thereby attenuating macromolecular crowding and

suppressing ribosome-driven liquid–liquid phase separation, which in turn reprograms tumor-associated macrophages from an immunosuppressive M2-like state to a pro-inflammatory M1-like phenotype. Concurrently, SN-38 induces immunogenic tumor cell death, establishing a synergistic chemo-immunotherapeutic loop that enhances antitumor efficacy and immunity. Notably, the SOP formulation not only improves drug solubility and tumor accumulation but also circumvents the systemic immunosuppression associated with free OSI-027, broadening its therapeutic window. By establishing the modulation of ribosome-driven LLPS as a previously unrecognized axis for immune reprogramming, this work proposes a conceptual advance that transcends conventional cytokine- or signaling-centric strategies. More broadly, it demonstrates that carrier-free nanoassemblies can function not merely as delivery vehicles but as active modulators of intracellular biophysics, offering a blueprint for future nanoplateforms that harness phase separation dynamics to potentiate cancer chemo-immunotherapy. Several important questions remain to be addressed, including the molecular identity of the principal LLPS-driving protein, the precise phase-separation-driving regions of the candidate proteins, and the broader applicability of the proposed axis to other tumor types and TAM subpopulations, which we are actively pursuing in ongoing studies.

MATERIALS AND METHODS

Materials, Cell Culture, and Animals

7-Ethyl-10-hydroxycamptothecin (SN-38), linoleic acid (LA), and Cell Counting Kit-8 (CCK8) were purchased from Sigma–Aldrich (MO, USA). The mTOR inhibitor (OSI-027) and Annexin V/propidium iodide (PI) apoptosis assay kits were purchased from MedChemExpress (NJ, USA). Antibodies (anti-Ki-67, anti-CD31, anti-CD163, anti-CD206, anti-CD16+CD32), 5-Ethynyl-2'-deoxyuridine (EdU) assay/EdU staining proliferation kits and TdT-mediated dUTP Nick-End Labeling (TUNEL) assay kits were purchased from Abcam (Cambridge, UK). All reagents used in cell culture were from Gibco (NY, USA). All other compounds and reagents were from Aladdin (Shanghai, China).

CFPAC-1, PANC-1 and THP-1 cells were purchased from the American Type Culture Collection (ATCC, USA). CFPAC-1 cells were cultured in high glucose DMEM supplemented with 10% fetal bovine serum (FBS). PANC-1 and THP-1 cells were cultured in RPMI-1640 (Roswell Park Memorial Institute 1640) containing 10% FBS. Both cell lines were maintained at 37 °C in 5% CO₂.

BALB/c nude mice with an average weight of 20 g (male, 6 weeks old) were purchased from the Model Animal Research Center of Nanjing University and maintained at the animal center of the First Affiliated Hospital, Zhejiang University School of Medicine. PDX specimens were also obtained from the First Affiliated Hospital, Zhejiang University School of Medicine. KPC (LSL-KrasG12D/+; LSL-Trp53R172H/+; Pdx1-Cre) mice were purchased from Jackson Laboratory and maintained at the animal center of Zhejiang Academy of Medical Sciences.

This study was performed in strict accordance with the NIH guidelines for the care and use of laboratory animals (NIH Publication No. 85e23 Rev. 1985) and was approved by the Institutional Animal Care and Use Committee at the First Affiliated Hospital of Zhejiang University (Ethics Approval No. 20241684) and Clinical Research Ethics Committee of the First Affiliated Hospital, Zhejiang University School of Medicine (Ethics Approval No. 20240006).

Preparation of SOP

LA-SN-38 was synthesized through a typical esterification reaction as reported previously.²⁵ Briefly, 236 mg of LA, 330 mg of SN-38, 178 mg of 1-(3-(dimethylamino)propyl)-3-ethylcarbodiimide hydrochlor-

ide (EDC-HCl), 113 mg of 4-dimethylaminopyridine (DMAP) and 164 μ L of N,N-diisopropylethylamine (DIEA) were dissolved in 8 mL of anhydrous N,N-dimethylformamide (DMF) and stirred at 25 °C overnight. Next, the reaction solution was treated with solvent removal, washing and purification to obtain LA-SN-38.

SN38 nano were prepared via a self-assembly method. The synthesized LA-SN-38 was first dissolved in DMSO and then quickly injected into ddH₂O with further ddH₂O dialysis.

OSI-027 was dissolved in DMSO in advance. SOPs were prepared by adding OSI-027 into the obtained SN-38 nano solution with a vortex. Additionally, further ddH₂O dialysis was adopted to remove DMSO, and lyophilization was performed.

Characterization of SOP

The synthesis of LA-SN-38 was confirmed via NMR (AVANCE III, Bruker, Switzerland) and UPLC-Triple-TOF/MS; Acquity (Waters, USA; 5600, AB SCIEX, USA). The solubility and stability of SN-38, LA-SN-38, OSI-027 and SOP in water were observed. HPLC (1260, Agilent, USA) was applied to analyze the EE and LC of OSI in SOP. The amount of OSI-027 was determined via a reverse-phase column (Agilent, HC-C18, 4.6 $\times$ 150 mm, 5 μ m) with a mobile phase of acetonitrile and water at 254 nm. EE/LC equal the ratio between the measured amount and OSI-027 added/SOP weighted.

Uniform suspensions of SOPs at concentrations ranging from 0.5 to 4 mg/mL were prepared. The particle size distribution and morphology of SOPs at each concentration were characterized using TEM (Hitachi S-4800). Samples were negatively stained with 2% (w/v) aqueous uranyl acetate solution and imaged under an accelerating voltage of 80 kV. Furthermore, the size and distribution of SOP in different solutions were measured by DLS (Nano ZS 90, Malvern, UK), and particle-size changes were also monitored over 3 days.

Cytotoxicity

Cell viability was measured with a CCK8 kit according to the manufacturer's protocol. CFPAC-1 and PANC-1 cells were seeded in 96-well plates at a density of 5000 cells/well and incubated overnight. To explore the best synergistic effect of drugs, the adherent cells were treated with different concentrations of SN-38 (0.1, 0.2, 0.5, 1, 2, or 5 μ M) and different ratios of SN-38 and OSI-027 (S/O = 1:1, 1:2, 1:3, 2:1 or 3:1). Cells treated with SN-38 or OSI-027 alone (S/O = 1:0 or 0:1) were set as the control group. After 48 h of incubation, CCK8 solution (10 μ L) was added to each well, and its optical density (OD) value was then detected at 450 nm after another 2 h of incubation by a microplate reader (SPECTRA max 384, Molecular Devices, USA). All experiments were performed in triplicate.

CCK8 and EdU assays were also applied to confirm the cytotoxicity of the different groups of drugs. PANC-1 and CFPAC-1 cells were treated with OSI-027, SN-38 nano, SN-38 nano + OSI-027 (SN + OSI) and SOP, which were prepared following the IC₅₀ and best ratio (S/O = 2:1) from the above CCK8 assay. A saline group was set as the blank control. The labeling process was carried out following the manufacturer's instructions, and the samples were visualized via fluorescence microscopy (NanoZoomer 2.0-RS, Hamamatsu, Japan).

Apoptosis

The ability of SN-38 and OSI-027 to induce apoptosis was assessed via an Alexa Fluor 488 Annexin V/PI double-staining assay by flow cytometry (BD Biosciences, NJ, USA). PANC-1 and CFPAC-1 cells (1 $\times$ 10⁶/well) were incubated with OSI-027, SN-38 nano, SN+OSI and SOP for 48 h. Then, the cells were collected and washed with cold PBS, followed by the addition of binding buffer (200 μ L) to prepare the cell suspension. Next, Annexin V (5 μ L) and PI (5 μ L) were added, and the mixed suspension was incubated in the dark for approximately 15 min at room temperature. Finally, binding buffer (300 μ L) was added, and the apoptotic rate was measured.

Cellular Uptake

Cells were plated in 24-well plates with 5 $\times$ 10⁴ cells per well 1 day prior to the uptake experiments. The cells were preincubated with various inhibitors, CPZ (25 μ M), M- β -CD (10 μ g/mL), indomethacin (50 μ M), filipin (5 μ g/mL), colchicine (25 μ M) and amiloride

(50 μ M), for 30 min, followed by the addition of SOP or SN+OSI (S/O = 2:1, IC50) and incubation for 6 h. The cells were then washed three times with PBS and collected. Afterward, DMSO was added to breakdown the cell structure and dissolve the drugs for cellular uptake, which was then quantified by HPLC.

Animal Experiments and Antitumor Efficacy

In this study, CDX, PDX and KPC models of PDAC were all built to investigate the synergetic effect of SN-38 and OSI-027 *in vivo*. PANC-1-cell (1×10^6) suspension was subcutaneously injected into the backs of mice to establish the CDX model. When mice bearing subcutaneous xenografts reached 50–100 mm³, mice were randomized into five groups and intravenously injected with saline, OSI-027, SN-38 nano, SN+OSI or SOP every other day for 14 days (OSI-027:10 mg/kg body weight; SN-38:20 mg/kg body weight). For the PDX model, 1 mm³ tumor tissues were removed from PDX rats and transferred to 20 rats, which were divided into five groups at random. Every tumor volume {volume = [length $\times$ (width)²/2]} and body weight were recorded every 2 days. After tumor volumes reached 50–100 mm³, OSI-027, SN-38 nano, SN+OSI and SOP were injected through the tail vein (OSI-027:10 mg/kg body weight; SN-38:20 mg/kg body weight), and an equal volume of saline was adopted as a blank control. The injection was carried out every 2 days for 2 weeks. Rats were euthanized by cervical dislocation 2 weeks later, and their xenografts were extracted, followed by measuring tumor volume and weight. The tumor regression rate was defined as the (mean tumor weight of the control group – mean tumor weight of the experimental group)/mean tumor weight of the control group $\times$ 100%. The changes in alanine aminotransferase (ALT), aspartate aminotransferase (AST), total bilirubin (TBIL) and creatinine (CREA) in serum were investigated by corresponding kits (Abcam, Cambridge, UK). KPC mice were obtained by hybridization and fed for three months. This model is widely regarded as one of the most robust and clinically relevant spontaneous models of PDAC, faithfully recapitulating key histopathological, molecular, and behavioral features of human disease. After being given the same dosage regimen, the mice were injected with ¹⁸F-FDG and subjected to imaging via a PET scanner (Siemens, Biograph16HR, Germany).

Histopathology Studies and TUNEL Assay

Immunohistochemical staining and TUNEL assays were performed using paraffin-embedded tumor sections. Ki-67 and CD31 expression were assessed by antibodies, and apoptotic signals were detected by a Fluorescein TUNEL Kit in accordance with the manufacturer's protocol. CD31-stained vessels, Ki-67-positive cells and TUNEL-stained cells were all visualized by a light microscope and quantitatively analyzed via ImageJ software, indicating tumor angiogenesis, proliferation *in vivo* and cell apoptosis *in situ*, respectively.

Quantification of Drug in Tumor Tissues

The concentrations of SN-38 and OSI-027 remaining in tumor tissues were calculated by HPLC analysis. After tissue grinding, the suspension was centrifuged at 3000 rpm for 5 min to obtain the upper solution for sample preparation. The amounts of SN-38 and OSI-027 were determined at 266 and 254 nm, respectively.

Western Blotting Assay

Western blotting analysis (WB) was performed to determine the levels of the resistance indices ABCG-2 and MDRI and the reference gene GAPDH. Briefly, RIPA lysis buffer containing 1 mM phenylmethanesulfonyl fluoride (PMSF) was added to tumor sections followed by grinding. The cells in tumor sites were subsequently extracted and harvested from the upper supernatant. Samples with equivalent amounts of protein were subjected to 10% SDS–polyacrylamide gel electrophoresis (PAGE) and then transferred to polyvinylidene difluoride (PVDF) membranes. After incubation with primary antibodies at 4 °C overnight, including ABCG-2, MDRI and GAPDH, horseradish peroxidase (HRP)-linked secondary antibody (Beyotime, China) was added at room temperature. The bands were visualized by chemiluminescence using ECL plus reagents (Amersham

Pharmacia Biotech, USA) and ImageQuant LAS 4000 (GE, USA). Meanwhile, PDAC cells *in vitro* were treated with OSI-027, SN-38, SN+OSI or SOP for 24 h, followed by the same steps described above.

Flow Cytometry Analysis

FACS was employed for cell sorting. The tumors were digested by type IV collagenase, hyaluronidase and DNA enzyme and then subjected to cell extraction. The extracted cells were first fixed using fixation/permeabilization solution, washed three times with perm wash buffer, and then incubated with CD163 and CD206 antibodies for flow cytometry analysis. For CD16/32 antibody staining, the extracted cells were blocked with anti-CD16/32 in advance. For cells from PBMCs *in vivo* and cells cultured *in vitro*, the procedure was started from fixation directly after treatment.

Culture System of THP-1 Cells with OSI-027 *In Vitro*

To explore the effect of OSI-027 on THP-1 cell differentiation, THP-1 cells were cultured separately and treated with OSI-027 for 24 h. After centrifugation, the supernatant liquid was collected, and the secretion of various cytokines was detected with an ELISA kit (R&D Systems, USA) following the kit instructions. The expression of differentiation markers (CD163, CD206, CD16/32) of substratum cells was analyzed by FACS.

Coculture System of PDAC and THP-1 Cells *In Vitro*

To construct the coculture system, a 12 mm transwell system with a 0.4 μ m pore polyester membrane (Corning, USA) was used. THP-1 cells were cocultured with PANC-1/CFPAC-1 cells in RPMI 1640 medium containing 10% fetal bovine serum, penicillin and streptomycin until the logarithmic phase, and the cells were taken and made into a cell suspension. PANC-1/CFPAC-1 cells were seeded on the lower compartment of transwell inserts, and THP-1 cells were seeded into the upper compartment. Subsequently, media with different groups of drugs were added. The THP-1 and PDAC cell coculture system without drug treatment was set as a control, while the simple THP-1-cell group was set as a blank. After 72 h, the levels of cytokines in the coculture system and antibodies against THP-1 cells were determined by ELISA and FACS, respectively.

Cell Apoptosis and Phagocytosis of the Coculture System *In Vitro*

For cell apoptosis and phagocytosis, the above coculture system was established again. OSI-027 was added to groups with or without SN-38-treated coculture systems, and the simple PDAC cell group was used as a blank for apoptosis and the simple THP-1-cell group for phagocytosis. After 72 h, the lower PDAC cell apoptosis was measured via an Annexin V-FITC/PI staining assay. Meanwhile, the upper THP-1 cells were harvested, followed by labeling with fluorescence-labeled latex beads (1 μ m diameter, Polysciences, USA) and washing. Finally, FACS was performed for analysis.

Coculture System of HUVECs and Supernatant Fluid from PDAC and THP-1 Cells

In addition, THP-1 cells were harvested from the above coculture system of PDAC and THP-1 cells. After another 48 h of single culture, the supernatant fluid was transferred to further culture with HUVECs. After 8 h, the microvasculature of the cell samples was observed under a microscope.

Ultrastructure of THP-1 Cells

The effect of OSI-027 on cellular ultrastructure was assessed by TEM. THP-1 cells were treated with OSI-027 and SOP for 24 h. Following fixation, dehydration, permeabilization, embedding, sectioning, and staining, the samples were prepared and the ultrastructural changes of the cells were observed under a TEM.

LLPS and mTOR Association Analysis

Acquisition of Data Set. NCBI GEO (<https://www.ncbi.nlm.nih.gov/>) (Gene Expression Omnibus, GEO) is a microarray and high-throughput sequencing database. As of April 27, 2025, the inclusion criteria are (1) Independent expression profiles of pancreatic

cancer; (2) Data sets containing normal and tumor group information; (3) Samples with the largest possible sample size, with a minimum total of 100 samples; (4) Tissue specimens from human *Homo sapiens*. According to these criteria, we downloaded GSE62452 (sequencing platform GPL6244 [HuGene-1_0-st] Affymetrix Human Gene 1.0 ST Array [trans normal ipt gene version]), which includes 69 tumor samples and 61 normal samples. We then downloaded the corresponding mRNA probe expression matrix files and annotation files for the sequencing platform. Probes were individually converted to gene symbols, with probes not matching gene symbols being removed. For probes mapping to the same gene, the average value was taken as the gene's expression level, resulting in a gene expression matrix for subsequent analysis. We collected publicly available literature from PubMed (<https://pubmed.ncbi.nlm.nih.gov/>) and four online databases PhaSepDB (<http://db.phasep.pro/>), LLPSDB (<http://bio-comp.org.cn>), PhaSePro (<https://phasepro.elte.hu/>), and DrLLPS (<http://llps.biocuckoo.cn>) to obtain LLPS data. Subsequently, we identified and deduplicated human genes for subsequent analysis. The HALLMARK_MTORC1_SIGNALING metabolic-related gene set was downloaded from the MSigDB database (<http://www.gseamsigdb.org/gsea/msigdb/index.jsp>).

Recognition of Macrophage M2. The CIBERSORT algorithm was used to calculate the level of macrophage M2 infiltration.⁵⁰ The Wilcoxon test was employed to examine the distribution differences in macrophage M2 infiltration levels between tumor and normal tissues.

Identify Macrophage M2-Related Gene Modules by WGCNA. The WGCNA method identifies gene clusters (or modules) in high-throughput data that exhibit strong correlations with sample phenotypes. It systematically extracts characteristic genes within these clusters and calculates modules with significant associations to phenotypic traits. The approach begins by assuming gene networks follow scale-free properties, defining gene coexpression correlation matrices and adjacency functions derived from network structures. By calculating pairwise dissimilarity coefficients between nodes, it identifies gene sets associated with specific phenotypic traits. Using the R package WGCNA (Version 1.72–5, <https://www.an.r-project.org/web/packages/WGCNA/>),⁵¹ we analyzed MAD (median absolute deviation)-top5000 gene expression data with macrophage M2 as the phenotype. The screening process involved three key steps: (1) Identifying outliers through hierarchical clustering; (2) Determining optimal soft threshold parameters within the 1–20 range using the “pick Soft Threshold” function; (3) Constructing a Topological Overlap Matrix (TOM) by converting the correlation matrix into an adjacency matrix and then transforming it into a topological overlap matrix. (4) Construct hierarchical clustering trees using average linkage (with optimal alignment of linked genes), then identify distinct gene modules through dynamic tree-cutting (minimum module size: 200). Merge similar modules based on predefined cut thresholds within each group. (5) Evaluate the correlation between gene modules and phenotypic traits using Pearson correlation coefficients.

Acquisition of Differentially Expressed Genes (DEGs)

Using the limma package (<https://bioconductor.org/packages/release/bioc/html/limma.html>) version 3.58.1 in R for differential analysis between tumor and normal tissues,⁵¹ we calculated p-values and logFC values for genes. Additionally, multiple testing was corrected using the Benjamini-Hochberg method to obtain adjusted p-values (adj.P.Value). We evaluated differences based on two criteria: adj.P.Value < 0.05 and $|\log_2\text{FC}| \geq 0.5$ (differential fold change of 1.5 times).

Identify Intersecting Genes

Venn diagram analysis was performed using online tools (<https://bioinfop.cnb.csic.es/tools/venny/>) to obtain the intersection of genes obtained by cross between Macrophages_M2, DEGs, LLPS and mTORC1.

Relevance Analysis

The intersection genes obtained by Venn diagram analysis using online tools (<https://bioinfop.cnb.csic.es/tools/venny/>) for Macro-

phages_M2, DEGs and LLPS genes were used for subsequent analysis. Based on the HALLMARK_MTORC1_SIGNALING gene set, we used the GSEA algorithm (<https://www.bioconductor.org/packages/release/bioc/html/GSEA.html>) in R to calculate mTORC1 signaling levels.⁵² Subsequently, we employed the Spearman function from the ggplot2 package (Version 3.5.0: <https://github.com/tidyverse/ggplot2>) to analyze the association between the aforementioned intersecting gene expressions and MTORC1 SIGNALING.

Statistical Analysis

The results are expressed as the mean $\pm$ SD (standard deviation). All data were obtained from at least three independent experiments. Statistical analysis was conducted using one-way ANOVA with post hoc tests. Multiple comparisons between the groups were performed using Tukey's test. Spearman correlation coefficients were used to evaluate correlations between two continuous variables. Categorical variables were compared using chi-square tests, while continuous variables were analyzed through Wilcoxon rank sum tests or *t* tests. All statistical tests were double-sided. Statistically significant differences were defined as **p* < 0.05, ***p* < 0.01, ****p* < 0.001 and *****p* < 0.0001.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.5c17081>.

Detailed experimental results, including the Raptor and Rictor genes analysis in PDAC (Figure S1); characterization of LA-SN38 (Figure S2) and OSI-027 entrapment in SOP (Figure S3); fundamental properties of SOP nanoparticles (Figure S4); *in vitro* cytotoxicity assessment (Figure S5); *in vivo* antitumor efficiency of SOP in CDX models (Figure S6); *in vivo* biosafety evaluation of SOP in CDX models (Figure S7); *in vivo* antitumor efficiency of SOP in KPC models (Figure S8); pharmacokinetic analysis (Figure S9); biodistribution (Figure S10); induction performance (Figure S11); immunomodulatory ability (Figures S12–S13); LLPS inhibition and analysis (Figures S14–S15) (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Wei Chen – Department of General Surgery, Sir Run Run Shaw Hospital, School of Medicine and Key Laboratory for accurate diagnosis and treatment of abdominal infection in Zhejiang province, Sir Run Run Shaw Hospital, School of Medicine, Zhejiang University, Zhejiang 310016, PR China; orcid.org/0000-0002-0399-6216; Email: wei_chen@zju.edu.cn

Xiaoling Xu – Shulan International Medical College, Zhejiang Shuren University, Hangzhou 310015, PR China; orcid.org/0000-0002-7472-0495; Email: ziyao1988@zju.edu.cn

Authors

Xiaoguang Wang – Department of Hepatobiliary Surgery, Affiliated Hospital of Jiaxing University, Jiaxing 314000, P.R. China

Lisha Cai – Cancer Institute of Integrated Traditional Chinese and Western Medicine, Zhejiang Key Laboratory of Disease-Syndrome Integration for Cancer Prevention and Treatment, Zhejiang Academy of Traditional Chinese Medicine, Tongde Hospital of Zhejiang Province, Hangzhou 310012, PR China

Xiaoxiao Zheng – Cancer Institute of Integrated Traditional Chinese and Western Medicine, Zhejiang Key Laboratory of Disease-Syndrome Integration for Cancer Prevention and Treatment, Zhejiang Academy of Traditional Chinese Medicine, Tongde Hospital of Zhejiang Province, Hangzhou 310012, PR China

Li Zheng – Department of General Surgery, Sir Run Run Shaw Hospital, School of Medicine, Zhejiang University, Zhejiang 310016, PR China

Hao Liu – Cancer Institute of Integrated Traditional Chinese and Western Medicine, Zhejiang Key Laboratory of Disease-Syndrome Integration for Cancer Prevention and Treatment, Zhejiang Academy of Traditional Chinese Medicine, Tongde Hospital of Zhejiang Province, Hangzhou 310012, PR China

Baoming Wang – Department of General Surgery, Sir Run Run Shaw Hospital, School of Medicine, Zhejiang University, Zhejiang 310016, PR China

Liqiang Hu – Key Laboratory for accurate diagnosis and treatment of abdominal infection in zhejiang province, Sir Run Run Shaw Hospital, School of Medicine, Zhejiang University, Zhejiang 310016, PR China

Shangzhi Xie – Cancer Institute of Integrated Traditional Chinese and Western Medicine, Zhejiang Key Laboratory of Disease-Syndrome Integration for Cancer Prevention and Treatment, Zhejiang Academy of Traditional Chinese Medicine, Tongde Hospital of Zhejiang Province, Hangzhou 310012, PR China

Shufen Zhang – Cancer Institute of Integrated Traditional Chinese and Western Medicine, Zhejiang Key Laboratory of Disease-Syndrome Integration for Cancer Prevention and Treatment, Zhejiang Academy of Traditional Chinese Medicine, Tongde Hospital of Zhejiang Province, Hangzhou 310012, PR China

Xuemei Lu – Key Laboratory for accurate diagnosis and treatment of abdominal infection in zhejiang province, Sir Run Run Shaw Hospital, School of Medicine, Zhejiang University, Zhejiang 310016, PR China

Jiayan Mao – Cancer Institute of Integrated Traditional Chinese and Western Medicine, Zhejiang Key Laboratory of Disease-Syndrome Integration for Cancer Prevention and Treatment, Zhejiang Academy of Traditional Chinese Medicine, Tongde Hospital of Zhejiang Province, Hangzhou 310012, PR China

Yiliang Lin – Department of Chemical and Biomolecular Engineering, National University of Singapore, Singapore 117585, Singapore; orcid.org/0000-0001-9403-9464

Xiang Gao – Shenzhen Institutes of Advanced Technology, Chinese Academy of Science, Shenzhen 518055, PR China; orcid.org/0000-0003-0699-0351

Jian Li – School of Bioscience and Technology, Chengdu Medical College, Chengdu 610500, PR China

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsnano.5c17081>

Author Contributions

X.W.: Conceptualization, Investigation, Funding acquisition. L.C.: Investigation, Visualization, Writing—original draft. X.Z.: Investigation, Visualization. L.Z.: Investigation. H.L.: Resources, Formal analysis. B.W.: Investigation. L.H.: Methodology. S.X.: Validation. S.Z.: Validation. X.L.: Writing—review and editing. J.M.: Visualization. Y.L.: Validation. X.G.: Validation. J.L.: Resources. W.C. and X.X.: Conceptualization, Super-

vision, Project administration, Funding acquisition. X.W., L.C., and X.Z. contributed equally.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors wish to express their gratitude for the financial support from the Zhejiang Provincial Natural Science Foundation of China (LR20H160001 to Wei Chen, LY21H160046 to Xiaoguang Wang), Key R&D Projects of Zhejiang Province (2020C03G5263593 to Wei Chen), Young Qihuang Scholar of the National Administration of Traditional Chinese Medicine (to Wei Chen), Zhejiang Provincial Ten Thousand Plan for Young Top Talents (to Wei Chen), Training Objects of Health Innovative Talents of Zhejiang Health (to Wei Chen and Hao Liu), Key Project Coconstructed by Zhejiang Province and Ministry (WKJ-ZJ-1916 to Wei Chen), the Natural Science Foundation of China (81972693, 82171827), and the Zhejiang-Germany International Joint Laboratory on Oxidative Damage Medicine (2025-GHSYS-001).

REFERENCES

- (1) Hussain, A.; Fareed, A. Personalized medicine in pancreatic cancer: Harnessing the potential of mRNA vaccines. *J. Genet. Eng. Biotechnol.* **2025**, *23*, 100469.
- (2) Yao, W.; Rose, J. L.; Wang, W.; Seth, S.; Jiang, H.; Taguchi, A.; Liu, J.; Yan, L.; Kapoor, A.; Hou, P.; Chen, Z.; Wang, Q.; Nezi, L.; Xu, Z.; Yao, J.; Hu, B.; Pettazzoni, P. F.; Ho, I. L.; Feng, N.; Ramamoorthy, V.; Jiang, S.; Deng, P.; Ma, G. J.; Den, P.; Tan, Z.; Zhang, S. X.; Wang, H.; Wang, Y. A.; Deem, A. K.; Fleming, J. B.; Carugo, A.; Heffernan, T. P.; Maitra, A.; Viale, A.; Ying, H.; Hanash, S.; DePinho, R. A.; Draetta, G. F. Syndecan 1 is a critical mediator of macropinocytosis in pancreatic cancer. *Nature* **2019**, *568*, 410–414.
- (3) Fitzgerald, A. A.; Wang, S.; Agarwal, V.; Marcisak, E. F.; Zuo, A.; Jablonski, S. A.; Loth, M.; Fertig, E. J.; MacDougall, J.; Zhukovsky, E.; et al. DPP inhibition alters the CXCR3 axis and enhances NK and CD8⁺ T cell infiltration to improve anti-PD1 efficacy in murine models of pancreatic ductal adenocarcinoma. *J. Immunother. Cancer* **2021**, *9*, No. e002837.
- (4) Li, J.; Dai, Y.; Wang, T.; Zhang, X.; Du, P.; Dong, Y.; Jiao, Z. Polyphenol-based pH-responsive nanoparticles enhance chemotherapeutic in pancreatic cancer. *J. Controlled Release* **2025**, *380*, 615–629.
- (5) Zhou, J.; Zhang, K.; Liu, C.; Long, J.; Gu, H.; Dong, H.; Wu, C.; Xu, L.; Wang, F.; Li, Q. Utidelone suppresses PDAC growth and enhances gemcitabine therapy by inducing immunogenic cell death. *iScience* **2025**, *28*, 112509.
- (6) Werba, G.; Weissinger, D.; Kawaler, E. A.; Zhao, E.; Kalfakakou, D.; Dhara, S.; Wang, L.; Lim, H. B.; Oh, G.; Jing, X.; et al. Single-cell RNA sequencing reveals the effects of chemotherapy on human pancreatic adenocarcinoma and its tumor microenvironment. *Nat. Commun.* **2023**, *14*, 797.
- (7) Tabe, S.; Takeuchi, K.; Aoshima, K.; Okumura, A.; Yamamoto, Y.; Yanagisawa, K.; Eto, R.; Matsuo, M.; Ueno, Y.; Konishi, T.; Furukawa, Y.; Yamaguchi, K.; Morinaga, S.; Miyagi, Y.; Ohtsuka, M.; Tanimizu, N.; Taniguchi, H. A pancreatic cancer organoid incorporating macrophages reveals the correlation between the diversity of tumor-associated macrophages and cancer cell survival. *Biomaterials* **2025**, *314*, 122838.
- (8) Burstyn-Cohen, T.; Maimon, A. TAM receptors, Phosphatidylinositol, inflammation, and Cancer. *Cell Commun. Signal.* **2019**, *17* (1), 156.
- (9) Kloosterman, D. J.; Akkari, L. Macrophages at the interface of the co-evolving cancer ecosystem. *Cell* **2023**, *186*, 1627–1651.

- (10) Zhuang, J.; Hai, Y.; Lu, X.; Sun, B.; Fan, R.; Zhang, B.; Wang, W.; Han, B.; Luo, L.; Yang, L.; et al. A Self-Assembled Metabolic Regulator Reprograms Macrophages to Combat Cytokine Storm and Boost Sepsis Immunotherapy. *Research* **2025**, *8*, 0663.
- (11) Huo, Y.; Zhang, H.; Sa, L.; Zheng, W.; He, Y.; Lyu, H.; Sun, M.; Zhang, L.; Shan, L.; Yang, A.; Wang, T. M1 polarization enhances the antitumor activity of chimeric antigen receptor macrophages in solid tumors. *J. Transl. Med.* **2023**, *21* (1), 225.
- (12) Liu, D.; Yang, J.; Cristea, I. M. Liquid-liquid phase separation in innate immunity. *Trends Immunol.* **2024**, *45*, 454–469.
- (13) Shang, Z.; Zhang, S.; Wang, J.; Zhou, L.; Zhang, X.; Billadeau, D. D.; Yang, P.; Zhang, L.; Zhou, F.; Bai, P.; Jia, D. TRIM25 predominately associates with anti-viral stress granules. *Nat. Commun.* **2024**, *15*, 4127.
- (14) Chen, S.; Lu, K.; Hou, Y.; You, Z.; Shu, C.; Wei, X.; Wu, T.; Shi, N.; Zhang, G.; Wu, J.; et al. YY1 complex in M2 macrophage promotes prostate cancer progression by upregulating IL-6. *J. Immunother. Cancer* **2023**, *11*, No. e006020.
- (15) Delarue, M.; Brittingham, G. P.; Pfeffer, S.; Surovtsev, I. V.; Pinglay, S.; Kennedy, K. J.; Schaffer, M.; Gutierrez, J. L.; Sang, D.; Poterewicz, G.; et al. mTORC1 Controls Phase Separation and the Biophysical Properties of the Cytoplasm by Tuning Crowding. *Cell* **2018**, *174* (2), 338–349.e20.
- (16) Chen, W.; Ma, T.; Shen, X. N.; Xia, X. F.; Xu, G. D.; Bai, X. L.; Liang, T. B. Macrophage-induced tumor angiogenesis is regulated by the TSC2-mTOR pathway. *Cancer Res.* **2012**, *72*, 1363–1372.
- (17) Guri, Y.; Colombi, M.; Dazert, E.; Hindupur, S. K.; Roszik, J.; Moes, S.; Jenoe, P.; Heim, M. H.; Riezman, L.; Riezman, H.; et al. mTORC2 Promotes Tumorigenesis via Lipid Synthesis. *Cancer Cell* **2017**, *32* (6), 807–823.e12.
- (18) Battaglion, S.; Benjamin, D.; Wälchli, M.; Maier, T.; Hall, M. N. mTOR substrate phosphorylation in growth control. *Cell* **2022**, *185*, 1814–1836.
- (19) Chen, J.; Liu, K.; Liu, Y.; Wang, X.; Zhang, Z. Targeting mTORC1/2 with OSI-027 inhibits proliferation and migration of keloid keratinocytes. *Exp. Dermatol.* **2019**, *28*, 270–275.
- (20) Chen, B.; Xu, M.; Zhang, H.; Xu, M. Z.; Wang, X. J.; Tang, Q. H.; Tang, J. Y. The Antipancreatic Cancer Activity of OSI-027, a Potent and Selective Inhibitor of mTORC1 and mTORC2. *DNA Cell Biol.* **2015**, *34*, 610–617.
- (21) Chen, Y.; Tang, S.; Cai, F.; Wan, Y. Strategies for Small Extracellular Vesicle-Based Cancer Immunotherapy. *Research* **2024**, *7*, 0421.
- (22) Ding, J.; Zhao, X.; Long, S.; Sun, W.; Du, J.; Fan, J.; Peng, X. A photoactivatable tumor-targeting in situ nanovaccine for large-volume tumor therapy. *Smart Mol.* **2025**, *3*, No. e70014.
- (23) Niu, G.; Zhao, L.; Wang, Y.; Jiang, Y. PDA/gold nanorod-based nanoparticles for synergistic genetic and photothermal combination therapy for cancer treatment. *ChemPhysmater* **2023**, *2*, 83–89.
- (24) Bahaar, H.; Reddy, S. G.; Kumar, B. S.; Prashanthi, K.; Murthy, A. Modified Layered Double Hydroxide – PEG Magneto-Sensitive Hydrogels with Suitable Ligno-Alginate Green Polymer Composite for Prolonged Drug Delivery Applications. *Eng. Sci.* **2023**, *24*, 914.
- (25) Wang, H.; Xie, H.; Wang, J.; Wu, J.; Ma, X.; Li, L.; Wei, X.; Ling, Q.; Song, P.; Zhou, L.; et al. Self-Assembling Prodrugs by Precise Programming of Molecular Structures that Contribute Distinct Stability, Pharmacokinetics, and Antitumor Efficacy. *Adv. Funct. Mater.* **2015**, *25* (31), 4956–4965.
- (26) Li, M.; Cai, J.; Gao, Y. Recent advances in stimulus-assisted nanoprecipitation for nanoparticle synthesis. *Curr. Opin. Colloid Interface Sci.* **2026**, *81*, 101972.
- (27) Moghimi, S. M.; Szebeni, J. Stealth liposomes and long circulating nanoparticles: Critical issues in pharmacokinetics, opsonization and protein-binding properties. *Prog. Lipid Res.* **2003**, *42*, 463–478.
- (28) Butreddy, P.; Heo, J.; Rampal, N.; Liu, T.; Liu, L.; Smith, W.; Zhang, X.; Prange, M. P.; Legg, B. A.; Schenter, G. K.; et al. Ion Correlations Decrease Particle Aggregation Rate by Increasing Hydration Forces at Interfaces. *ACS Nano* **2024**, *18*, 26047–26055.
- (29) Sun, H.; Jiao, R.; An, G.; Xu, H.; Wang, D. Influence of particle size on the aggregation behavior of nanoparticles: Role of structural hydration layer. *J. Environ. Sci.* **2021**, *103*, 33–42.
- (30) Kim, W.; Ly, N. K.; He, Y.; Li, Y.; Yuan, Z.; Yeo, Y. Protein corona: Friend or foe? Co-opting serum proteins for nanoparticle delivery. *Adv. Drug Delivery Rev.* **2023**, *192*, 114635.
- (31) Kopac, T. Protein corona, understanding the nanoparticle-protein interactions and future perspectives: A critical review. *Int. J. Biol. Macromol.* **2021**, *169*, 290–301.
- (32) Ahmad, A.; Khan, J. M.; Paray, B. A.; Rashid, K.; Parvez, A. Endolysosomal trapping of therapeutics and endosomal escape strategies. *Drug Discovery Today* **2024**, *29*, 104070.
- (33) Wang, X.; Li, H.; Chen, C.; Liang, Z. Understanding of endo/lysosomal escape of nanomaterials in biomedical application. *Smart Mol.* **2025**, *3*, No. e20240017.
- (34) Zhao, S.; Yu, N.; Han, H.; Guo, S.; Murthy, N. Advances in acid-degradable and enzyme-cleavable linkers for drug delivery. *Curr. Opin. Chem. Biol.* **2025**, *84*, 102552.
- (35) Fang, J.; Islam, W.; Maeda, H. Exploiting the dynamics of the EPR effect and strategies to improve the therapeutic effects of nanomedicines by using EPR effect enhancers. *Adv. Drug Delivery Rev.* **2020**, *157*, 142–160.
- (36) Zhi, X.; Chen, W.; Xue, F.; Liang, C.; Chen, B. W.; Zhou, Y.; Wen, L.; Hu, L.; Shen, J.; Bai, X.; et al. OSI-027 inhibits pancreatic ductal adenocarcinoma cell proliferation and enhances the therapeutic effect of gemcitabine both in vitro and in vivo. *Oncotarget* **2015**, *6* (28), 26230–26241.
- (37) Sherman, M. H.; Beatty, G. L. Tumor Microenvironment in Pancreatic Cancer Pathogenesis and Therapeutic Resistance. *Annu. Rev. Pathol.* **2023**, *18*, 123–148.
- (38) Mills, C. D. M1 and M2 Macrophages: Oracles of Health and Disease. *Crit. Rev. Immunol.* **2012**, *32*, 463–488.
- (39) Yunna, C.; Mengru, H.; Lei, W.; Weidong, C. Macrophage M1/M2 polarization. *Eur. J. Pharmacol.* **2020**, *877*, 173090.
- (40) Wang, P.; Wang, H.; Huang, Q.; Peng, C.; Yao, L.; Chen, H.; Qiu, Z.; Wu, Y.; Wang, L.; Chen, W. Exosomes from M1-Polarized Macrophages Enhance Paclitaxel Antitumor Activity by Activating Macrophages-Mediated Inflammation. *Theranostics* **2019**, *9*, 1714–1727.
- (41) Boutilier, A. J.; ElSawa, S. F. Macrophage Polarization States in the Tumor Microenvironment. *Int. J. Mol. Sci.* **2021**, *22*, 6995.
- (42) An, H.; Ordureau, A.; Körner, M.; Paulo, J. A.; Harper, J. W. Systematic quantitative analysis of ribosome inventory during nutrient stress. *Nature* **2020**, *583*, 303–309.
- (43) Liu, X.; Shao, Y.; Li, Y.; Chen, Z.; Shi, T.; Tong, Q.; Zou, X.; Ju, L.; Pan, J.; Zhuang, R.; Pan, X. Extensive Review of Nanomedicine Strategies Targeting the Tumor Microenvironment in PDAC. *Int. J. Nanomed.* **2025**, *20*, 3379–3406.
- (44) Hezaveh, K.; Shinde, R. S.; Klötgen, A.; Halaby, M. J.; Lamorte, S.; Ciudad, M. T.; Quevedo, R.; Neufeld, L.; Liu, Z. Q.; Jin, R.; et al. Tryptophan-derived microbial metabolites activate the aryl hydrocarbon receptor in tumor-associated macrophages to suppress anti-tumor immunity. *Immunity* **2022**, *55* (2), 324–340.e8.
- (45) Cao, Y.; Wo, M.; Xu, C.; Fei, X.; Jin, J.; Shan, Z. An AMPK agonist suppresses the progress of colorectal cancer by regulating the polarization of TAM to M1 through inhibition of HIF-1 α and mTOR signal pathway. *J. Cancer Res. Ther.* **2023**, *19*, 1560–1567.
- (46) Wenes, M.; Shang, M.; Di Matteo, M.; Goveia, J.; Martín-Pérez, R.; Serneels, J.; Prenen, H.; Ghesquière, B.; Carmeliet, P.; Mazzone, M. Macrophage Metabolism Controls Tumor Blood Vessel Morphogenesis and Metastasis. *Cell Metab.* **2016**, *24*, 701–715.
- (47) Guo, R.; Wang, R.; Zhang, W.; Li, Y.; Wang, Y.; Wang, H.; Li, X.; Song, J. Macrophage Polarisation in the Tumour Microenvironment: Recent Research Advances and Therapeutic Potential of Different Macrophage Reprogramming. *Cancer Control* **2025**, *32*, 10732748251316604.
- (48) Liu, R.; Shen, Y.; Wei, Y.; Zhu, C.; Yao, X.; Gu, X.; Qi, L.; Sun, H. The central role of macrophages in skeletal muscle regeneration:

Phenotypic Polarization, metabolic reprogramming, and therapeutic prospects. *Biochem. Pharmacol.* **2026**, 250, 118025.

(49) Ma, X.; Yang, S.; Zhang, T.; Wang, S.; Yang, Q.; Xiao, Y.; Shi, X.; Xue, P.; Kang, Y.; Liu, G.; Sun, Z. J.; Xu, Z. Bioresponsive immune-booster-based prodrug nanogel for cancer immunotherapy. *Acta Pharm. Sin. B* **2022**, 12, 451–466.

(50) Liu, S.; Wang, Z.; Zhu, R.; Wang, F.; Cheng, Y.; Liu, Y. Three Differential Expression Analysis Methods for RNA Sequencing: limma, EdgeR, DESeq2. *J. Visualized Exp.* **2021**, No. e62528.

(51) Langfelder, P.; Horvath, S. WGCNA: An R package for weighted correlation network analysis. *BMC Bioinf.* **2008**, 9, 559.

(52) Xiao, B.; Liu, L.; Li, A.; Xiang, C.; Wang, P.; Li, H.; Xiao, T. Identification and Verification of Immune-Related Gene Prognostic Signature Based on ssGSEA for Osteosarcoma. *Front. Oncol.* **2020**, 10, 607622.